

The $n \times n$ matrix C is invertible if and only if $\text{rank}(C) = n$ —that is, if and only if the column vectors of C form a basis for n -space. In this case, the criterion $AC = CD$ in Theorem 5.2 can be written as $D = C^{-1}AC$. The equation $D = C^{-1}AC$ transforms a matrix A into a diagonal matrix D that is much easier to work with, as we will see in a moment. We give a formal definition of such a transformation of A into D , followed by a corollary summarizing the results of this paragraph.

DEFINITION 5.3 Diagonalizable Matrix

An $n \times n$ matrix A is **diagonalizable** if there exists an invertible matrix C such that $C^{-1}AC = D$, a diagonal matrix. The matrix C is said to **diagonalize** the matrix A .

COROLLARY 1 A Criterion for Diagonalization

An $n \times n$ matrix A is diagonalizable if and only if n -space has a basis consisting of eigenvectors of A .

If an $n \times n$ matrix A is diagonalizable as described in Definition 5.3, then we have $A = CDC^{-1}$ as well. From this we obtain

$$A^k = \underbrace{(CDC^{-1})(CDC^{-1})(CDC^{-1}) \cdots (CDC^{-1})}_{k \text{ factors}} \quad (1)$$

The adjacent terms $C^{-1}C$ cancel in Eq. (1) to give $A^k = CD^kC^{-1}$. Thus, the computation of A^k is essentially reduced to the computation of D^k . We summarize these observations as a corollary to Theorem 5.2.

COROLLARY 2 Computation of A^k

Let an $n \times n$ matrix A have n eigenvectors and eigenvalues, giving rise to the matrices C and D so that $AC = CD$, as described in Theorem 5.2. If the n eigenvectors are independent, then C is an invertible matrix and $C^{-1}AC = D$. Under these conditions, we have

$$A^k = CD^kC^{-1}$$

Diagonalization of square matrices plays a very important role in linear algebra. As we show in the next theorem, diagonalization of an $n \times n$ matrix A can always be achieved if the characteristic polynomial has n distinct roots.

THEOREM 5.3 Independence of Eigenvectors

Let A be an $n \times n$ matrix. If $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ are eigenvectors of A corresponding to *distinct* eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$, respectively, the set $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is linearly independent and A is diagonalizable.

PROOF Suppose that the conclusion is false, so the eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ are linearly dependent. Then one of them is a linear combination of its predecessors. (See Exercise 37, page 203.) Let $\mathbf{v}_k$ be the first such vector, so that

$$\mathbf{v}_k = d_1 \mathbf{v}_1 + d_2 \mathbf{v}_2 + \cdots + d_{k-1} \mathbf{v}_{k-1} \quad (2)$$

and $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{k-1}\}$ is independent. Multiplying Eq. (2) by λ_k , we obtain

$$\lambda_k \mathbf{v}_k = d_1 \lambda_k \mathbf{v}_1 + d_2 \lambda_k \mathbf{v}_2 + \cdots + d_{k-1} \lambda_k \mathbf{v}_{k-1}. \quad (3)$$

On the other hand, multiplying both sides of Eq. (2) on the left by the matrix A yields

$$\lambda_k \mathbf{v}_k = d_1 \lambda_1 \mathbf{v}_1 + d_2 \lambda_2 \mathbf{v}_2 + \cdots + d_{k-1} \lambda_{k-1} \mathbf{v}_{k-1}, \quad (4)$$

because $A\mathbf{v}_i = \lambda_i \mathbf{v}_i$. Subtracting Eq. (4) from Eq. (3), we see that

$$0 = d_1(\lambda_k - \lambda_1)\mathbf{v}_1 + d_2(\lambda_k - \lambda_2)\mathbf{v}_2 + \cdots + d_{k-1}(\lambda_k - \lambda_{k-1})\mathbf{v}_{k-1}.$$

This last equation is a dependence relation because not all the coefficients are zero. (Not all d_i are zero because of Eq. (2) and because the λ_i are distinct.) But this contradicts the linear independence of the set $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{k-1}\}$. We conclude that $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is independent. That A is diagonalizable follows at once from Corollary 1 of Theorem 5.2. $\blacktriangle$

EXAMPLE 1 Diagonalize the matrix $A = \begin{bmatrix} -3 & 5 \\ -2 & 4 \end{bmatrix}$, and compute A^k in terms of k .

SOLUTION We compute

$$\det(A - \lambda I) = \begin{vmatrix} -3 - \lambda & 5 \\ -2 & 4 - \lambda \end{vmatrix} = \lambda^2 - \lambda - 2 = (\lambda - 2)(\lambda + 1).$$

The eigenvalues of A are $\lambda_1 = 2$ and $\lambda_2 = -1$. For $\lambda_1 = 2$, we have

$$A - 2I = \begin{bmatrix} -5 & 5 \\ -2 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix},$$

which yields an eigenvector

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

For $\lambda_2 = -1$, we have

$$A + I = \begin{bmatrix} -2 & 5 \\ -2 & 5 \end{bmatrix} \sim \begin{bmatrix} 2 & -5 \\ 0 & 0 \end{bmatrix},$$

which yields an eigenvector

$$\mathbf{v}_2 = \begin{bmatrix} 5 \\ 2 \end{bmatrix}.$$

A diagonalization of A is given by

$$A = CDC^{-1} = \begin{bmatrix} 1 & 5 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} -\frac{2}{3} & \frac{5}{3} \\ \frac{1}{3} & -\frac{1}{3} \end{bmatrix}.$$

(We omit the computation of C^{-1} .) Thus,

$$A^k = \begin{bmatrix} 1 & 5 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 2^k & 0 \\ 0 & (-1)^k \end{bmatrix} \begin{bmatrix} -\frac{2}{3} & \frac{5}{3} \\ \frac{1}{3} & -\frac{1}{3} \end{bmatrix} = \frac{1}{3} \begin{bmatrix} -2^{k+1} \pm 5 & 5(2^k) \mp 5 \\ -2^{k+1} \pm 2 & 5(2^k) \mp 2 \end{bmatrix},$$

the colored sign being used only when k is odd. ■

Diagonalization has uses other than computing A^k . The next section presents an application to differential equations. Section 8.1 gives another application. Here is an application to geometry in $\mathbb{R}^2$.

EXAMPLE 2 Find a formula for the linear transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ that reflects vectors in the line $x + 2y = 0$.

SOLUTION We saw in Illustration 2 of Section 5.1 that T has eigenvectors $[1, 2]$ and $[2, -1]$ with corresponding eigenvalues -1 and 1 , respectively. Let A be the standard matrix representation of T . Using column-vector notation, we have

$$T\left(\begin{bmatrix} 1 \\ 2 \end{bmatrix}\right) = A\begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} -1 \\ -2 \end{bmatrix} \quad \text{and} \quad T\left(\begin{bmatrix} 2 \\ -1 \end{bmatrix}\right) = A\begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}.$$

Letting $C = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$ and $D = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$, we see that $AC = CD$ as in Theorem 5.2. Because C is invertible,

$$A = CDC^{-1} = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{5} & \frac{2}{5} \\ \frac{2}{5} & -\frac{1}{5} \end{bmatrix} = \frac{1}{5} \begin{bmatrix} 3 & -4 \\ -4 & -3 \end{bmatrix}.$$

Thus

$$T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = A\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \frac{1}{5} \begin{bmatrix} 3x - 4y \\ -4x - 3y \end{bmatrix}$$

or, in row notation, $T([x, y]) = \frac{1}{5}[3x - 4y, -4x - 3y]$. ■

The preceding matrices A and D provide examples of *similar matrices*, a term we now define.

DEFINITION 5.4 Similar Matrices

An $n \times n$ matrix P is **similar** to an $n \times n$ matrix Q if there exists an invertible $n \times n$ matrix C such that $C^{-1}PC = Q$.

The relationship “ P is similar to Q ” satisfies the properties required for an *equivalence relation* (see Exercise 16). In particular, if P is similar to Q , then Q is also similar to P . This means that similarity need not be stated in a directional way; we can simply say that P and Q are similar matrices.

EXAMPLE 3 Find a diagonal matrix similar to the matrix

$$A = \begin{bmatrix} 1 & 0 & 0 \\ -8 & 4 & -6 \\ 8 & 1 & 9 \end{bmatrix}$$

of Example 6 in Section 5.1.

SOLUTION Taking $r = s = t = 1$ in Example 6 of Section 5.1, we see that eigenvalues and corresponding eigenvectors of A are given by

$$\lambda_1 = 1, \quad \lambda_2 = 6, \quad \lambda_3 = 7,$$

$$\mathbf{v}_1 = \begin{bmatrix} 15 \\ 8 \\ -16 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ -3 \\ 1 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix}.$$

If we let

$$C = \begin{bmatrix} 15 & 0 & 0 \\ 8 & -3 & -2 \\ -16 & 1 & 1 \end{bmatrix},$$

then Theorem 5.3 tells us that C is invertible. Theorem 5.2 then shows that

$$C^{-1}AC = D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 7 \end{bmatrix}.$$

HISTORICAL NOTE THE IDEA OF SIMILARITY, like many matrix notions, appears without a definition in works from as early as the 1820s. In fact, in his 1826 work on quadratic forms (see note on page 409), Cauchy showed that if two quadratic forms (polynomials) are related by a change of variables—that is, if their matrices are similar—then their characteristic equations are the same. But like the concept of orthogonality, that of similarity was first formally defined and discussed by Georg Frobenius in 1878. Frobenius began by discussing the general case: he called two matrices A, D *equivalent* if there existed invertible matrices P, Q such that $D = PAQ$. The latter matrices were called the *substitutions* through which A was transformed into D .

Frobenius then dealt with the special cases where $P = Q^T$ (the two matrices were then called *congruent*) and where $P = Q^{-1}$ (the similarity case of this section). Frobenius went on to prove many results on similarity, including the useful theorem that, if A is similar to D , then $f(A)$ is similar to $f(D)$, where f is any polynomial matrix function.

Thus D is similar to A . We are not eager to check that $C^{-1}AC = D$; however, it is easy to check the equivalent statement:

$$AC = CD = \begin{bmatrix} 15 & 0 & 0 \\ 8 & -18 & -14 \\ -16 & 6 & 7 \end{bmatrix}.$$

It is not always essential that a matrix have distinct eigenvalues in order to be diagonalizable. As long as the $n \times n$ matrix A has n *independent* eigenvectors to form the column vectors of an invertible C , we have $C^{-1}AC = D$, the diagonal matrix of the eigenvalues corresponding to C .

EXAMPLE 4 Diagonalize the matrix

$$A = \begin{bmatrix} 1 & -3 & 3 \\ 0 & -5 & 6 \\ 0 & -3 & 4 \end{bmatrix}.$$

SOLUTION We find that the characteristic equation of A is

$$\begin{aligned} (1 - \lambda)((-5 - \lambda)(4 - \lambda) + 18) &= (1 - \lambda)(\lambda^2 + \lambda - 2) \\ &= (1 - \lambda)(\lambda + 2)(\lambda - 1) = 0. \end{aligned}$$

Thus, the eigenvalues of A are $\lambda_1 = 1$, $\lambda_2 = 1$, and $\lambda_3 = -2$. Notice that 1 is a root of multiplicity 2 of the characteristic equation; we say that the eigenvalue 1 has *algebraic multiplicity* 2.

Reducing $A - I$, we obtain

$$A - I = \begin{bmatrix} 0 & -3 & 3 \\ 0 & -6 & 6 \\ 0 & -3 & 3 \end{bmatrix} \sim \begin{bmatrix} 0 & 1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

We see that the eigenspace E_1 (that is, the nullspace of $A - I$) has dimension 2 and consists of vectors of the form

$$\begin{bmatrix} s \\ r \\ r \end{bmatrix} \text{ for any scalars } r \text{ and } s.$$

Taking $s = 1$ and $r = 0$, and then taking $s = 0$ and $r = 1$, we obtain the independent eigenvectors

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad \text{and} \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

corresponding to the eigenvalues $\lambda_1 = \lambda_2 = 1$.

Reducing $A + 2I$, we find that

$$A + 2I = \begin{bmatrix} 3 & -3 & 3 \\ 0 & -3 & 6 \\ 0 & -3 & 6 \end{bmatrix} \sim \begin{bmatrix} 3 & 0 & -3 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{bmatrix}.$$

Thus an eigenvector corresponding to $\lambda_3 = -2$ is

$$v_3 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}.$$

Therefore, if we take

$$C = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & 1 & 1 \end{bmatrix},$$

we should have

$$C^{-1}AC = D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{bmatrix}.$$

A check shows that indeed

$$AC = CD = \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & -4 \\ 0 & 1 & -2 \end{bmatrix}.$$

As we indicated in Example 4, the **algebraic multiplicity** of an eigenvalue λ_i of A is its multiplicity as a root of the characteristic equation of A . Its **geometric multiplicity** is the dimension of the eigenspace E_{λ_i} . Of course, the geometric multiplicity of each eigenvalue must be at least 1, because there always exists a nonzero eigenvector in the eigenspace. However, it is possible for the algebraic multiplicity to be greater than the geometric multiplicity.

EXAMPLE 5 Referring back to Examples 4 and 5 in Section 5.1, find the algebraic and geometric multiplicities of the eigenvalue 2 of the matrix

$$A = \begin{bmatrix} 2 & 1 & 0 \\ -1 & 0 & 1 \\ 1 & 3 & 1 \end{bmatrix}.$$

SOLUTION Example 4 on page 292 shows that the characteristic equation of A is $-(\lambda - 2)^2(\lambda + 1) = 0$, so 2 is an eigenvalue of algebraic multiplicity 2. Example 5 on page 293 shows that the reduced form of $A - 2I$ is

$$\begin{bmatrix} 1 & 3 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Thus, the eigenspace E_2 , which is the set of vectors of the form

$$\begin{bmatrix} r \\ 0 \\ r \end{bmatrix} \text{ for } r \in \mathbb{R},$$

has dimension 1, so the eigenvalue 2 has geometric multiplicity 1. ■

We state a relationship between the algebraic multiplicity and the geometric multiplicity of a (possibly complex) eigenvalue. (See Exercise 33 in Section 9.4.)

The geometric multiplicity of an eigenvalue of a matrix A is less than or equal to its algebraic multiplicity.

Let $\lambda_1, \lambda_2, \dots, \lambda_m$ be the distinct (possibly complex) eigenvalues of an $n \times n$ matrix A . Let B_i be a basis for the eigenspace of λ_i for $i = 1, 2, \dots, m$. It can be shown by an argument similar to the proof of Theorem 5.3 that the union of these bases B_i is an independent set of vectors in n -space (see Exercise 24). Corollary 1 on page 307 shows that the matrix A is diagonalizable if this union of the B_i is a basis for n -space. This will occur precisely when the geometric multiplicity of each eigenvalue is equal to its algebraic multiplicity. (See the boxed statement.) Conversely, it can be shown that, if A is diagonalizable, the algebraic multiplicity of each eigenvalue is the same as its geometric multiplicity. We summarize this in a theorem.

THEOREM 5.4 A Criterion for Diagonalization

An $n \times n$ matrix A is diagonalizable if and only if the algebraic multiplicity of each (possibly complex) eigenvalue is equal to its geometric multiplicity.

Thus the 3×3 matrix in Example 4 is diagonalizable, because its eigenvalue 1 has algebraic and geometric multiplicity 2, and its eigenvalue -2 has algebraic and geometric multiplicity 1. However, the matrix in Example 5 is not diagonalizable, because the eigenvalue 2 has algebraic multiplicity 2 but geometric multiplicity 1.

In Section 9.4, we show that every square matrix A is similar to a matrix J , its *Jordan canonical form*. If A is diagonalizable, then J is a diagonal matrix, found precisely as in the preceding examples. If A is not diagonalizable, then J again has the eigenvalues of A on its main diagonal, but it also has entries 1 immediately above some of the diagonal entries. The remaining entries are all zero. For example, the matrix

$$A = \begin{bmatrix} 2 & 1 & 0 \\ -1 & 0 & 1 \\ 1 & 3 & 1 \end{bmatrix}$$

of Example 5 has Jordan canonical form

$$J = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix}.$$

Section 9.4 describes a technique for finding J . This Jordan canonical form is as close to a diagonalization of A as we can come. The Jordan canonical form has applications to the solution of systems of differential equations.

To conclude this discussion, we state a result whose proof requires an excursion into complex numbers. The proof is given in Chapter 9. Thus far, we have seen nothing to indicate that symmetric matrices play any significant role in linear algebra. The following theorem immediately elevates them into a position of prominence.

THEOREM 5.5 Diagonalization of Real Symmetric Matrices

Every real symmetric matrix is *real diagonalizable*. That is, if A is an $n \times n$ symmetric matrix with real-number entries, then each eigenvalue of A is a real number, and its algebraic multiplicity equals its geometric multiplicity.

A diagonalizing matrix C for a symmetric matrix A can be chosen to have some very nice properties, as we will show for real symmetric matrices in Chapter 6.

SUMMARY

Let A be an $n \times n$ matrix.

1. If A has n distinct eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$, and C is an $n \times n$ matrix having as j th column vector an eigenvector corresponding to λ_j , then $C^{-1}AC$ is the diagonal matrix having λ_j on the main diagonal in the j th column.
2. If $C^{-1}AC = D$, then $A^k = CD^kC^{-1}$.
3. A matrix P is similar to a matrix Q if there exists an invertible matrix C such that $C^{-1}PC = Q$.
4. The algebraic multiplicity of an eigenvalue λ of A is its multiplicity as a root of the characteristic equation; its geometric multiplicity is the dimension of the corresponding eigenspace E_λ .
5. Any eigenvalue's geometric multiplicity is less than or equal to its algebraic multiplicity.
6. The matrix A is diagonalizable if and only if the geometric multiplicity of each of its eigenvalues is the same as the algebraic multiplicity.
7. Every symmetric matrix is diagonalizable. All eigenvalues of a real symmetric matrix are real numbers.

EXERCISES

In Exercises 1–8, find the eigenvalues λ_i and the corresponding eigenvectors $\mathbf{v}_i$ of the given matrix A , and also find an invertible matrix C and a diagonal matrix D such that $D = C^{-1}AC$.

1. $A = \begin{bmatrix} -3 & 4 \\ 4 & 3 \end{bmatrix}$

2. $A = \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}$

3. $A = \begin{bmatrix} 7 & 8 \\ -4 & -5 \end{bmatrix}$

4. $A = \begin{bmatrix} 6 & 3 & -3 \\ -2 & -1 & 2 \\ 16 & 8 & -7 \end{bmatrix}$

5. $A = \begin{bmatrix} -3 & 10 & -6 \\ 0 & 7 & -6 \\ 0 & 0 & 1 \end{bmatrix}$

6. $A = \begin{bmatrix} -3 & 5 & -20 \\ 2 & 0 & 8 \\ 2 & 1 & 7 \end{bmatrix}$

7. $A = \begin{bmatrix} -2 & 0 & -1 \\ 0 & 2 & 0 \\ 3 & 0 & 2 \end{bmatrix}$

8. $A = \begin{bmatrix} -4 & 6 & -12 \\ 3 & -1 & 6 \\ 3 & -3 & 8 \end{bmatrix}$

In Exercises 9–12, determine whether the given matrix is diagonalizable.

9. $\begin{bmatrix} 1 & 2 & 6 \\ 2 & 0 & -4 \\ 6 & -4 & 3 \end{bmatrix}$

10. $\begin{bmatrix} 3 & 1 & 0 \\ 0 & 3 & 1 \\ 0 & 0 & 3 \end{bmatrix}$

11. $\begin{bmatrix} -1 & 4 & 2 & -7 \\ 0 & 5 & -3 & 6 \\ 0 & 0 & -5 & 1 \\ 0 & 0 & 0 & 11 \end{bmatrix}$

12. $\begin{bmatrix} 3 & 2 & 5 & 1 \\ 2 & 0 & 2 & 6 \\ 5 & 2 & 7 & -1 \\ 1 & 6 & -1 & 3 \end{bmatrix}$

13. Mark each of the following True or False.

- a. Every $n \times n$ matrix is diagonalizable.
- b. If an $n \times n$ matrix has n distinct real eigenvalues, it is diagonalizable.
- c. Every $n \times n$ real symmetric matrix is real diagonalizable.
- d. An $n \times n$ matrix is diagonalizable if and only if it has n distinct eigenvalues.
- e. An $n \times n$ matrix is diagonalizable if and only if the algebraic multiplicity of each of its eigenvalues equals the geometric multiplicity.
- f. Every invertible matrix is diagonalizable.
- g. Every triangular matrix is diagonalizable.
- h. If A and B are similar square matrices and A is diagonalizable, then B is also diagonalizable.

- i. If an $n \times n$ matrix A is diagonalizable, there is a unique diagonal matrix D that is similar to A .
- j. If A and B are similar square matrices, then $\det(A) = \det(B)$.

14. Give two different diagonal matrices that are similar to the matrix $\begin{bmatrix} 1 & 4 \\ 0 & -3 \end{bmatrix}$.

15. Prove that, if a matrix is diagonalizable, so is its transpose.
16. Let P , Q , and R be $n \times n$ matrices. Recall that P is similar to Q if there exists an invertible $n \times n$ matrix C such that $C^{-1}PC = Q$. This exercise shows that similarity is an *equivalence relation*.
- a. (Reflexive.) Show that P is similar to itself.
 - b. (Symmetric.) Show that, if P is similar to Q , then Q is similar to P .
 - c. (Transitive.) Show that, if P is similar to Q and Q is similar to R , then P is similar to R .

17. Prove that, for every square matrix A all of whose eigenvalues are real, the product of its eigenvalues is $\det(A)$.

18. Prove that similar square matrices have the same eigenvalues with the same algebraic multiplicities.

19. Let A be an $n \times n$ matrix.

- a. Prove that if A is similar to rA where r is a real scalar other than 1 or -1 , then all eigenvalues of A are zero. [HINT: See the preceding exercise.]
- b. What can you say about A if it is diagonalizable and similar to rA for some r where $|r| \neq 1$?
- c. Find a nonzero 2×2 matrix A which is similar to rA for every $r \neq 0$. (See part a.)
- d. Show that $A = \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix}$ is similar to $-A$.
(Observe that the eigenvalues of A are not all zero.)

20. Let A be a real tridiagonal matrix—that is, an $n \times n$ matrix for $n > 2$ all of whose entries are zero except possibly those of the

form $a_{i,i-1}$, a_{ii} , or $a_{i,i+1}$. Show that if $a_{i,i-1}$ and $a_{i,i+1}$ are both positive, both negative, or both zero for $i = 2, 3, \dots, n$, then A has real eigenvalues. [HINT: Show that a diagonal matrix D can be found such that DAD^{-1} is real symmetric.]

21. Find a formula for the linear transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ that reflects vectors in the line $y = mx$. [HINT: Proceed as in Example 2.]
22. Let A and C be $n \times n$ matrices, and let C be invertible. Prove that, if v is an eigenvector of A with corresponding eigenvalue λ , then $C^{-1}v$ is an eigenvector of $C^{-1}AC$ with corresponding eigenvalue λ . Then prove that all eigenvectors of $C^{-1}AC$ are of the form $C^{-1}v$, where v is an eigenvector of A .
23. Explain how we can deduce from Exercise 22 that, if A and B are similar square matrices, each eigenvalue of A has the same geometric multiplicity for A that it has for B . (See Exercise 18 for the corresponding statement on algebraic multiplicities.)
24. Prove that, if $\lambda_1, \lambda_2, \dots, \lambda_m$ are distinct real eigenvalues of an $n \times n$ real matrix A and if B_i is a basis for the eigenspace E_{λ_i} , then the union of the bases B_i is an independent set of vectors in $\mathbb{R}^n$. [HINT: Make use of Theorem 5.3.]
25. Let $T: V \rightarrow V$ be a linear transformation of a vector space V into itself. Prove that, if $v_1, v_2, \dots, v_k$ are eigenvectors of T corresponding to distinct nonzero eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_k$, then the set $\{T(v_1), T(v_2), \dots, T(v_k)\}$ is independent.
26. Show that the set $\{e^{\lambda_1 x}, e^{\lambda_2 x}, \dots, e^{\lambda_k x}\}$, where the λ_i are distinct, is independent in the vector space W of all functions mapping $\mathbb{R}$ into $\mathbb{R}$ and having derivatives of all orders.
27. Using Exercise 26, show that the infinite set $\{e^{kx} \mid k \in \mathbb{R}\}$ is an independent set in the vector space W described in Exercise 26. [HINT: How many vectors are involved in any dependence relation?]
28. In Section 5.1, we stated that if we allow complex numbers, then an $n \times n$ matrix with entries chosen at random has eigenvectors that form a basis for n -space with probability 1. In light of our work in

this section, give the best justification you can for this statement.

 29. The MATLAB command

$$A = (2*r)*\text{rand}(n) - r*\text{ones}(n); \text{eig}(A)$$

should exhibit the eigenvalues of an $n \times n$ matrix A with random entries between $-r$ and r . Illustrate your argument in the preceding exercise for seven matrices with values of r from 1 to 50 and values of n from 2 to 8. (Recall that we used $\text{rand}(n)$ in the MATLAB exercises of Section 1.6.)

In Exercises 30–41, use the routines *MATCOMP* and *ALLROOTS* in *IINTEK* or use *MATLAB* to classify the given matrix as real diagonalizable, complex (but not real) diagonalizable, or not diagonalizable. (Load the matrices from a matrix file if it is accessible.)

$$30. \begin{bmatrix} 18 & 25 & -25 \\ 1 & 6 & -1 \\ 18 & 34 & -25 \end{bmatrix} \quad 31. \begin{bmatrix} 8.3 & 8.0 & -6.0 \\ -2.0 & 0.3 & 3.0 \\ 0.0 & 0.0 & 4.3 \end{bmatrix}$$

$$32. \begin{bmatrix} 24.55 & 46.60 & 46.60 \\ -4.66 & -8.07 & -9.32 \\ -9.32 & -18.64 & -17.39 \end{bmatrix}$$

$$33. \begin{bmatrix} 0.8 & -1.6 & 1.8 \\ -0.6 & -3.8 & 3.4 \\ -20.6 & -1.2 & 9.6 \end{bmatrix}$$

$$34. \begin{bmatrix} 7 & -20 & -5 & 5 \\ 5 & -13 & -5 & 0 \\ -5 & 10 & 7 & 5 \\ 5 & -10 & -5 & -3 \end{bmatrix}$$

$$35. \begin{bmatrix} 2 & 5 & -9 & 10 \\ 4 & 9 & 8 & -3 \\ 8 & 2 & 0 & 12 \\ 7 & -6 & 3 & 2 \end{bmatrix}$$

$$36. \begin{bmatrix} -22.7 & -26.9 & -6.3 & -46.5 \\ -59.7 & -40.9 & 20.9 & -99.5 \\ 15.9 & 9.6 & -8.4 & 26.5 \\ 43.8 & 36.5 & -7.3 & 78.2 \end{bmatrix}$$

$$37. \begin{bmatrix} 66.2 & 58.0 & -11.6 & 116.0 \\ 120.6 & 89.6 & -42.6 & 201.0 \\ -21.0 & -15.0 & 7.6 & -35.0 \\ -99.6 & -79.0 & 28.6 & -169.4 \end{bmatrix}$$

$$38. \begin{bmatrix} -253 & -232 & -96 & 1088 & 280 \\ 213 & 204 & 93 & -879 & -225 \\ -90 & -90 & -47 & 360 & 90 \\ -38 & -36 & -18 & 162 & 40 \\ 62 & 64 & 42 & -251 & -57 \end{bmatrix}$$

$$39. \begin{bmatrix} 154 & -24 & -36 & -1608 & -336 \\ -126 & 16 & 18 & 1314 & 270 \\ 54 & 0 & 4 & -540 & -108 \\ 24 & 0 & 0 & -236 & -48 \\ -42 & -12 & -18 & 366 & 70 \end{bmatrix}$$

$$40. \begin{bmatrix} -2513 & 596 & -414 & -2583 & 1937 \\ 127 & -32 & 33 & 132 & -81 \\ -421 & 94 & -83 & -434 & 306 \\ 2610 & -615 & 443 & 2684 & -1994 \\ 90 & -19 & 29 & 94 & -50 \end{bmatrix}$$

$$41. \begin{bmatrix} 2 & -4 & 6 & 2 & 1 \\ 6 & 3 & -8 & 1 & 2 \\ -4 & 0 & 5 & 1 & 2 \\ 4 & 3 & 5 & -11 & 3 \\ 6 & 7 & 9 & 2 & 3 \end{bmatrix}$$

5.3

TWO APPLICATIONS

In this section, A will always denote a matrix with real-number entries.

In Section 5.1, we were motivated to introduce eigenvalues by our desire to compute $A^k \mathbf{x}$. Recall that we regard $\mathbf{x}$ as an initial information vector of some process, and A as a matrix that transforms, by left multiplication, an information vector at any stage of the process into the information vector at the next stage. In our first application, we examine this computation of $A^k \mathbf{x}$ and the significance of the eigenvalues of A . As illustration, we determine the behavior of the terms F_n of the Fibonacci sequence for large values of n . Our second application shows how diagonalization of matrices can be used to solve some linear systems of differential equations.

Application: Computing $A^k \mathbf{x}$

Let A be a real-diagonalizable $n \times n$ matrix, and let $\lambda_1, \lambda_2, \dots, \lambda_n$ be the n not necessarily distinct eigenvalues of A . That is, each eigenvalue of A is repeated in this list in accord with its algebraic multiplicity. Let $B = (\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)$ be an ordered basis for $\mathbb{R}^n$, where $\mathbf{v}_j$ is an eigenvector for λ_j . We have seen that, if C is the matrix having $\mathbf{v}_j$ as j th column vector, then

$$C^{-1}AC = D = \begin{bmatrix} \lambda_1 & & & 0 \\ & \lambda_2 & & \\ & & \ddots & \\ 0 & & & \lambda_n \end{bmatrix}$$

For any vector $\mathbf{x}$ in $\mathbb{R}^n$, let $\mathbf{d} = [d_1, d_2, \dots, d_n]$ be its coordinate vector relative to the basis B . Thus,

$$\mathbf{x} = d_1 \mathbf{v}_1 + d_2 \mathbf{v}_2 + \cdots + d_n \mathbf{v}_n.$$

Then

$$A^k \mathbf{x} = d_1 \lambda_1^k \mathbf{v}_1 + d_2 \lambda_2^k \mathbf{v}_2 + \cdots + d_n \lambda_n^k \mathbf{v}_n. \quad (1)$$

Equation (1) expresses $A^k \mathbf{x}$ as a linear combination of the eigenvectors $\mathbf{v}_j$.

Let us regard $\mathbf{x}$ as an initial *information vector* in a process in which the information vector at the next stage of the process is found by multiplying the present information vector on the left by a matrix A . Illustrations of this situation are provided by Markov chains in Section 1.7 and the generation of the Fibonacci sequence in Section 5.1. We are interested here in the long-term outcome of the process. That is, we wish to study $A^k \mathbf{x}$ for large values of k .

Let us number our eigenvalues and eigenvectors in Eq. (1) so that $|\lambda_i| \geq |\lambda_j|$ if $i < j$; that is, the eigenvalues are arranged in order of decreasing magnitude. Suppose that $|\lambda_1| > |\lambda_2|$ so that λ_1 is the unique eigenvalue of maximum magnitude. Equation (1) may be written

$$A^k \mathbf{x} = \lambda_1^k (d_1 \mathbf{v}_1 + d_2 (\lambda_2/\lambda_1)^k \mathbf{v}_2 + \cdots + d_n (\lambda_n/\lambda_1)^k \mathbf{v}_n).$$

Thus, if k is large and $d_1 \neq 0$, the vector $A^k \mathbf{x}$ is approximately equal to $d_1 \lambda_1^k \mathbf{v}_1$ in the sense that $\|A^k \mathbf{x} - d_1 \lambda_1^k \mathbf{v}_1\|$ is small compared with $\|A^k \mathbf{x}\|$.

EXAMPLE 1 Show that a diagonalizable transition matrix T for a Markov chain has no eigenvalues of magnitude > 1 .

SOLUTION Example 3 in Section 5.1 shows that 1 is an eigenvalue for every transition matrix of a Markov chain. For every choice of population distribution vector $\mathbf{p}$, the vector $T^k \mathbf{p}$ is again a vector with nonnegative entries having sum 1. The preceding discussion shows that all eigenvalues of T must have magnitude ≤ 1 ; otherwise, entries in some $T^k \mathbf{p}$ would have very large magnitude as k increases. ■

EXAMPLE 2 Find the order of magnitude of the term F_k of the Fibonacci sequence $F_0, F_1, F_2, F_3, \dots$ —that is, the sequence

$$0, 1, 1, 2, 3, 5, 8, 13, \dots$$

for large values of k .

SOLUTION We saw in Section 5.1 that, if we let

$$\mathbf{x}_k = \begin{bmatrix} F_k \\ F_{k-1} \end{bmatrix},$$

then

$$\mathbf{x}_k = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \mathbf{x}_{k-1}.$$

We compute relation (1) for

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \quad \text{and} \quad \mathbf{x} = \mathbf{x}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

The characteristic equation of A is

$$(1 - \lambda)(-\lambda) - 1 = \lambda^2 - \lambda - 1 = 0.$$

Using the quadratic formula, we find the eigenvalues

$$\lambda_1 = \frac{1 + \sqrt{5}}{2} \quad \text{and} \quad \lambda_2 = \frac{1 - \sqrt{5}}{2}.$$

Reducing $A - \lambda_1 I$, we obtain

$$A - \lambda_1 I \sim \begin{bmatrix} \frac{1 - \sqrt{5}}{2} & 1 \\ 0 & 0 \end{bmatrix}, \quad \text{so} \quad \mathbf{v}_1 = \begin{bmatrix} 2 \\ \sqrt{5} - 1 \end{bmatrix}$$

is an eigenvector for λ_1 . In an analogous fashion, we find that

$$\mathbf{v}_2 = \begin{bmatrix} -2 \\ \sqrt{5} + 1 \end{bmatrix}$$

is an eigenvector corresponding to λ_2 . Thus we take

$$C = \begin{bmatrix} 2 & -2 \\ \sqrt{5} - 1 & \sqrt{5} + 1 \end{bmatrix}.$$

To find the coordinate vector $\mathbf{d}$ of $\mathbf{x}_1$ relative to the basis $(\mathbf{v}_1, \mathbf{v}_2)$, we observe that $\mathbf{x}_1 = C\mathbf{d}$. We find that

$$C^{-1} = \frac{1}{4\sqrt{5}} \begin{bmatrix} \sqrt{5} + 1 & 2 \\ 1 - \sqrt{5} & 2 \end{bmatrix};$$

thus,

$$\begin{bmatrix} d_1 \\ d_2 \end{bmatrix} = C^{-1}\mathbf{x}_1 = C^{-1} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{4\sqrt{5}} \begin{bmatrix} \sqrt{5} + 1 \\ 1 - \sqrt{5} \end{bmatrix}.$$

Equation 1 takes the form

$$\begin{aligned} A^k \mathbf{x}_1 = \begin{bmatrix} F_{k+1} \\ F_k \end{bmatrix} &= \left(\frac{\sqrt{5} + 1}{4\sqrt{5}} \right) \left(\frac{1 + \sqrt{5}}{2} \right)^k \begin{bmatrix} 2 \\ \sqrt{5} - 1 \end{bmatrix} \\ &\quad - \left(\frac{\sqrt{5} - 1}{4\sqrt{5}} \right) \left(\frac{1 - \sqrt{5}}{2} \right)^k \begin{bmatrix} -2 \\ \sqrt{5} + 1 \end{bmatrix}. \quad (2) \end{aligned}$$

For large k , the k th power of the eigenvalue $\lambda_1 = (1 + \sqrt{5})/2$ dominates, so $A^k \mathbf{x}_1$ is approximately equal to the shaded portion of Eq. (2). Computing the second component of $A^k \mathbf{x}_1$, we find that

$$F_k = \frac{1}{\sqrt{5}} \left(\left(\frac{1 + \sqrt{5}}{2} \right)^k - \left(\frac{1 - \sqrt{5}}{2} \right)^k \right). \quad (3)$$

Thus,

$$F_k \approx \frac{1}{\sqrt{5}} \left(\frac{1 + \sqrt{5}}{2} \right)^k \quad \text{for large } k. \quad (4)$$

Indeed, because $|\lambda_2| = |(1 - \sqrt{5})/2| < 1$, we see that the contribution from this second eigenvalue to the right-hand side of Eq. (3) approaches zero as k increases. Because $|\lambda_2|^k/\sqrt{5} < \frac{1}{2}$ for $k = 1$ and hence for all k , we see that F_k can be characterized as the closest integer to $(1/\sqrt{5})((1 + \sqrt{5})/2)^k$ for all k . Approximation (4) verifies that F_k increases exponentially with k , as expected for a population of rabbits. ■

Example 2 is a typical analysis of a process in which an information vector after the k th stage is equal to $A^k \mathbf{x}_1$ for an initial information vector $\mathbf{x}_1$ and a diagonalizable matrix A . An important consideration is whether any of the eigenvalues of A have magnitude greater than 1. When this is the case, the components of the information vector may grow exponentially in magnitude, as illustrated by the Fibonacci sequence in Example 2, where $|\lambda_1| > 1$. On the other hand, if all the eigenvalues have magnitude less than 1, the components of the information vector must approach zero as k increases.

The process just described is called *unstable* if A has an eigenvalue of magnitude greater than 1, *stable* if all eigenvalues have magnitude less than 1, and *neutrally stable* if the maximum magnitude of the eigenvalues is 1. Thus a Markov chain is a neutrally stable process, whereas generation of the Fibonacci sequence is unstable. The eigenvectors are called the **normal modes** of the process.

In the type of process just described, we study information at evenly spaced time intervals. If we study such a process as the number of time intervals increases and their duration approaches zero, we find ourselves in calculus. Eigenvalues and eigenvectors play an important role in applications of calculus, especially in studying any sort of vibration. In these applications of calculus, components of vectors are functions of time. Our second application illustrates this.

Application: Systems of Linear Differential Equations

In calculus, we see the importance of the differential equation

$$\frac{dx}{dt} = kx$$

in simple rate of growth problems involving the time derivative of the amount x present of a single quantity. We may also write this equation as $x' = kx$, where we understand that x is a function of the time variable t . In more complex growth situations, n quantities may be present in amounts $x_1, x_2, \dots, x_n$. The rate of change of x_i may depend not only on the amount of x_i present, but also on the amounts of the other $n - 1$ quantities present at time t . We

consider a situation in which the rate of growth of each x_i depends linearly on the amounts present of the n quantities. This leads to a system of linear differential equations

$$\begin{aligned}x_1' &= a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \\x_2' &= a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n \\&\vdots \\x_n' &= a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n\end{aligned}\tag{5}$$

where each x_i is a differentiable function of the real variable t and each a_{ij} is a scalar. The simplest such system is the single differential equation

$$x' = ax,\tag{6}$$

which has the general solution

$$x(t) = ke^{at},\tag{7}$$

where k is a scalar. (See Example 8 on page 298.) Direct computation verifies that function (7) is the general solution of Eq. (6).

Turning to the solution of system (5), we write it in matrix form as

$$\mathbf{x}' = A\mathbf{x},\tag{8}$$

where

$$\mathbf{x} = \begin{bmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{bmatrix}, \quad \mathbf{x}' = \begin{bmatrix} x_1'(t) \\ x_2'(t) \\ \vdots \\ x_n'(t) \end{bmatrix}, \quad \text{and} \quad A = [a_{ij}].$$

If the matrix A is a diagonal matrix so that $a_{ij} = 0$ for $i \neq j$, then system (8) reduces to a system of n equations, each like Eq. (6), namely:

$$\begin{aligned}x_1' &= a_{11}x_1 \\x_2' &= a_{22}x_2 \\&\vdots \\x_n' &= a_{nn}x_n.\end{aligned}\tag{9}$$

The general solution is given by

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} k_1 e^{a_{11}t} \\ k_2 e^{a_{22}t} \\ \vdots \\ k_n e^{a_{nn}t} \end{bmatrix}.$$

In the general case, we try to diagonalize A and reduce system (8) to a system like (9). If A is diagonalizable, we have

$$D = C^{-1}AC = \begin{bmatrix} \lambda_1 & & & 0 \\ & \lambda_2 & & \\ & & \ddots & \\ 0 & & & \lambda_n \end{bmatrix}$$

for some invertible $n \times n$ matrix C . If we substitute $A = CDC^{-1}$, Eq. (8) takes the form $C^{-1}\mathbf{x}' = D(C^{-1}\mathbf{x})$, or

$$\mathbf{y}' = D\mathbf{y}, \quad (10)$$

where

$$\mathbf{x} = C\mathbf{y}. \quad (11)$$

(If we let $\mathbf{x} = C\mathbf{y}$, we can confirm that $\mathbf{x}' = C\mathbf{y}'$.) The general solution of system (10) is

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} k_1 e^{\lambda_1 t} \\ k_2 e^{\lambda_2 t} \\ \vdots \\ k_n e^{\lambda_n t} \end{bmatrix},$$

where the λ_j are eigenvalues of the diagonalizable matrix A . The general solution $\mathbf{x}$ of system (8) is then obtained from Eq. (11), using the corresponding eigenvectors $\mathbf{v}_j$ of A as columns of the matrix C .

We emphasize that we have described the general solution of system (8) only in the case where the matrix A is diagonalizable. The algebraic multiplicity of each eigenvalue of A must equal its geometric multiplicity. The following example illustrates that the eigenvalues need not be distinct, as long as their algebraic and geometric multiplicities are the same.

EXAMPLE 3 Solve the linear differential system

$$\begin{aligned} x_1' &= x_1 - x_2 - x_3 \\ x_2' &= -x_1 + x_2 - x_3 \\ x_3' &= -x_1 - x_2 + x_3. \end{aligned}$$

SOLUTION The first step is to diagonalize the matrix

$$A = \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & -1 \\ -1 & -1 & 1 \end{bmatrix}.$$

Expanding the determinant $|A - \lambda I|$ across the first row, we obtain

$$\begin{aligned} |A - \lambda I| &= \begin{vmatrix} 1 - \lambda & -1 & -1 \\ -1 & 1 - \lambda & -1 \\ -1 & -1 & 1 - \lambda \end{vmatrix} \\ &= (1 - \lambda) \begin{vmatrix} 1 - \lambda & -1 \\ -1 & 1 - \lambda \end{vmatrix} + \begin{vmatrix} -1 & -1 \\ -1 & 1 - \lambda \end{vmatrix} - \begin{vmatrix} -1 & 1 - \lambda \\ -1 & -1 \end{vmatrix} \\ &= (1 - \lambda)(\lambda^2 - 2\lambda) + 2(\lambda - 2) \\ &= (\lambda - 2)((1 - \lambda)\lambda + 2) \\ &= -(\lambda + 1)(\lambda - 2)^2. \end{aligned}$$

This yields eigenvalues $\lambda_1 = -1$ and $\lambda_2 = \lambda_3 = 2$.

Next we compute eigenvectors. For $\lambda_1 = -1$, we have

$$A + I = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & -2 \\ 0 & 3 & -3 \\ 0 & 0 & 0 \end{bmatrix},$$

which gives the eigenvector

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

For $\lambda_2 = \lambda_3 = 2$, we have

$$A - 2I = \begin{bmatrix} -1 & -1 & -1 \\ -1 & -1 & -1 \\ -1 & -1 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

which gives the independent eigenvectors

$$\mathbf{v}_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \quad \text{and} \quad \mathbf{v}_3 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}.$$

A diagonalizing matrix is then

$$C = \begin{bmatrix} 1 & -1 & -1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix},$$

so that

$$D = C^{-1}AC = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}.$$

Setting $\mathbf{x} = C\mathbf{y}$, we see that the system becomes

$$\begin{aligned} y_1' &= -y_1 \\ y_2' &= 2y_2 \\ y_3' &= 2y_3, \end{aligned}$$

whose solution is given by

$$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} k_1 e^{-t} \\ k_2 e^{2t} \\ k_3 e^{2t} \end{bmatrix}$$

Therefore, the solution to the original system is

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 & -1 & -1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} k_1 e^{-t} - k_2 e^{2t} - k_3 e^{2t} \\ k_1 e^{-t} + k_2 e^{2t} \\ k_1 e^{-t} + k_3 e^{2t} \end{bmatrix}.$$

In Section 9.4, we show that every square matrix A is similar to a matrix J in *Jordan canonical form*, so that $J = C^{-1}AC$. The Jordan canonical form J and the invertible matrix C may have complex entries. In the Jordan canonical form J , all nondiagonal entries are zero except for some entries 1 immediately above the diagonal entries. If the Jordan form $J = C^{-1}AC$ is known, the substitution $\mathbf{x} = C\mathbf{y}$ again reduces a system $\mathbf{x}' = A\mathbf{x}$ of linear differential equations to one that can be solved easily for $\mathbf{y}$. The solution of the original system is computed as $\mathbf{x} = C\mathbf{y}$, just as was the solution in Example 3. This is an extremely important technique in the study of differential equations.

SUMMARY

1. Let A be diagonalizable by a matrix C , let $\mathbf{x}$ be any column vector, and let $\mathbf{d} = C^{-1}\mathbf{x}$. Then

$$A^k \mathbf{x} = d_1 \lambda_1^k \mathbf{v}_1 + d_2 \lambda_2^k \mathbf{v}_2 + \cdots + d_n \lambda_n^k \mathbf{v}_n,$$

where $\mathbf{v}_j$ is the j th column vector of C , as described on page 318.

2. Let multiplication of a column information vector by A give the information vector for the next stage of a process, as described on page 320. The process is stable, neutrally stable, or unstable, according as the maximum magnitude of the eigenvalues of A is less than 1, equal to 1, or greater than 1, respectively.
3. The system $\mathbf{x}' = A\mathbf{x}$ of linear differential equations can be solved, if A is diagonalizable, using the following three steps.

Step 1. Find a matrix C so that $D = C^{-1}AC$ is a diagonal matrix.

Step 2. Solve the simpler diagonal system $\mathbf{y}' = D\mathbf{y}$.

Step 3. The solution of the original system is $\mathbf{x} = C\mathbf{y}$.

EXERCISES

- Let the sequence $a_0, a_1, a_2, \dots$ be given by $a_0 = 0, a_1 = 1$, and $a_k = (a_{k-1} + a_{k-2})/2$ for $k \geq 2$.
 - Find the matrix A that can be used to generate this sequence as we used a matrix to generate the Fibonacci sequence in Section 5.1.
 - Classify this generation process as stable, neutrally stable, or unstable.
 - Compute expression (1) for $x = \begin{bmatrix} a_1 \\ a_0 \end{bmatrix}$ for this process. Check computations with the first few terms of the sequence.
 - Use the answer to part (c) to estimate a_k for large k .
- Repeat Exercise 1 if $a_k = a_{k-1} - \left(\frac{3}{16}\right)a_{k-2}$ for $k \geq 2$.
- Repeat Exercise 1, but change the initial data to $a_0 = 1, a_1 = 0$.
- Repeat Exercise 1 if $a_k = \left(\frac{1}{2}\right)a_{k-1} + \left(\frac{3}{16}\right)a_{k-2}$ for $k \geq 2$.
- Repeat Exercise 1 if $a_k = a_{k-1} + \left(\frac{3}{4}\right)a_{k-2}$ for $k \geq 2$.

In Exercises 6–13, solve the given system of linear differential equations as outlined in the summary.

- $$\begin{aligned} x_1' &= 3x_1 - 5x_2 \\ x_2' &= 2x_2 \end{aligned}$$
- $$\begin{aligned} x_1' &= x_1 + 4x_2 \\ x_2' &= 3x_1 \end{aligned}$$
- $$\begin{aligned} x_1' &= x_1 + 2x_2 \\ x_2' &= 2x_1 + x_2 \end{aligned}$$
- $$\begin{aligned} x_1' &= 2x_1 + 2x_2 \\ x_2' &= x_1 + 3x_2 \end{aligned}$$
- $$\begin{aligned} x_1' &= 6x_1 + 3x_2 - 3x_3 \\ x_2' &= -2x_1 - x_2 + 2x_3 \\ x_3' &= 16x_1 + 8x_2 - 7x_3 \end{aligned}$$
- $$\begin{aligned} x_1' &= -3x_1 + 10x_2 - 6x_3 \\ x_2' &= 7x_2 - 6x_3 \\ x_3' &= x_3 \end{aligned}$$
- $$\begin{aligned} x_1' &= -3x_1 + 5x_2 - 20x_3 \\ x_2' &= 2x_1 + 8x_3 \\ x_3' &= 2x_1 + x_2 + 7x_3 \end{aligned}$$
- $$\begin{aligned} x_1' &= -2x_1 - x_3 \\ x_2' &= 2x_2 \\ x_3' &= 3x_1 + 2x_3 \end{aligned}$$

ORTHOGONALITY

We are accustomed to working in $\mathbb{R}^2$ with coordinates of vectors relative to the standard ordered basis $(\mathbf{e}_1, \mathbf{e}_2) = (\mathbf{i}, \mathbf{j})$. These basis vectors are orthogonal (perpendicular) and have length 1. The vectors in the standard ordered basis $(\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n)$ of $\mathbb{R}^n$ have these same two properties. It is precisely these two properties of the standard bases that make computations using coordinates relative to them quite easy. For example, computations of angles and of length are easy when we use coordinates relative to standard bases. Throughout linear algebra, computations using coordinates relative to bases consisting of orthogonal unit vectors are generally the easiest to perform, and they generate less error in work with a computer.

This chapter is devoted to orthogonality. Orthogonal projection, which is the main tool for finding a basis of orthogonal vectors, is developed in Section 6.1. Section 6.2 shows that every finite-dimensional inner-product space has a basis of orthogonal unit vectors, and shows how to construct such a basis from any given basis. Section 6.3 deals with orthogonal matrices and orthogonal linear transformations. In Section 6.4 we show that orthogonal projection can be achieved by matrix multiplication. To conclude the chapter, we give applications to the method of least squares and overdetermined linear systems in Section 6.5.

6.1

PROJECTIONS

The Projection of $\mathbf{b}$ on $\text{sp}(\mathbf{a})$

For convenience, we develop the ideas in this section for the vector space $\mathbb{R}^n$, using the dot product. The work is valid in general finite-dimensional inner-product spaces, using the more cumbersome notation $\langle \mathbf{u}, \mathbf{v} \rangle$ for the inner product of $\mathbf{u}$ and $\mathbf{v}$. An illustration in function spaces, using an inner product defined by an integral, appears at the end of this section.

ILLUSTRATION 1

A practical concern in vector applications involves determining what portion of a vector $\mathbf{b}$ can be considered to act in the direction given by another vector $\mathbf{a}$. A force vector acting in a certain direction may be moving a body along a line having a different direction. For example, suppose that you are trying to roll your stalled car off the road by pushing on the door jamb at the side, so you can reach in and control the steering wheel when necessary. You are not applying the force in precisely the direction in which the car moves, as you would be if you could push from directly behind the car. Such considerations lead the physicist to consider the projection $\mathbf{p}$ of the force vector $\mathbf{F}$ on a direction vector $\mathbf{a}$, as shown in Figure 6.1. In Figure 6.1, the vector $\mathbf{p}$ is found by dropping a perpendicular from the tip of $\mathbf{F}$ to the vector $\mathbf{a}$. ■

Figure 6.2 shows the situation where the projection of $\mathbf{F}$ on $\mathbf{a}$ has a direction opposite to the direction of $\mathbf{a}$; in terms of our car in Illustration 1, you would be applying the force to move the car backward rather than forward. Figure 6.2 suggests that it is preferable to speak of *projection on the subspace* $\text{sp}(\mathbf{a})$ (which in this example is a line) than to speak of *projection on* $\mathbf{a}$.

We derive geometrically a formula for the projection $\mathbf{p}$ of the force vector $\mathbf{F}$ on $\text{sp}(\mathbf{a})$, based on Figures 6.1 and 6.2. We see that $\mathbf{p}$ is a multiple of $\mathbf{a}$. Now $(1/\|\mathbf{a}\|)\mathbf{a}$ is a unit vector having the same direction as $\mathbf{a}$, so $\mathbf{p}$ is a scalar multiple of this unit vector. We need only find the appropriate scalar. Referring to Figure 6.1, we see that the appropriate scalar is $\|\mathbf{F}\| \cos \theta$, because it is the length of the leg labeled $\mathbf{p}$ of the right triangle. This same formula also gives the appropriate negative scalar for the case shown in Figure 6.2, because $\cos \theta$ is negative for this angle θ lying between 90° and 180° . Thus we obtain

$$\mathbf{p} = \frac{\|\mathbf{F}\| \cos \theta}{\|\mathbf{a}\|} \mathbf{a} = \frac{\|\mathbf{F}\| \|\mathbf{a}\| \cos \theta}{\|\mathbf{a}\| \|\mathbf{a}\|} \mathbf{a} = \frac{\mathbf{F} \cdot \mathbf{a}}{\mathbf{a} \cdot \mathbf{a}} \mathbf{a}.$$

Of course, we assume that $\mathbf{a} \neq \mathbf{0}$. We replace our force vector $\mathbf{F}$ by a general vector $\mathbf{b}$ and box this formula.

Projection $\mathbf{p}$ of $\mathbf{b}$ on $\text{sp}(\mathbf{a})$ in $\mathbb{R}^n$

$$\mathbf{p} = \frac{\mathbf{b} \cdot \mathbf{a}}{\mathbf{a} \cdot \mathbf{a}} \mathbf{a} \quad (1)$$

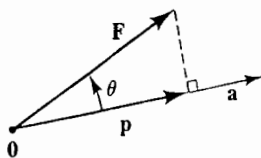


FIGURE 6.1
Projection $\mathbf{p}$ of $\mathbf{F}$ on $\mathbf{a}$.

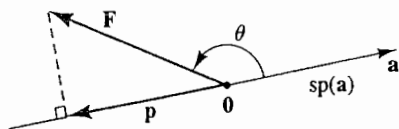


FIGURE 6.2
Projection $\mathbf{p}$ of $\mathbf{F}$ on $\text{sp}(\mathbf{a})$.

EXAMPLE 1 Find the projection $\mathbf{p}$ of the vector $[1, 2, 3]$ on $\text{sp}\{[2, 4, 3]\}$ in $\mathbb{R}^3$.

SOLUTION We let $\mathbf{a} = [2, 4, 3]$ and $\mathbf{b} = [1, 2, 3]$ in formula (1), obtaining

$$\mathbf{p} = \frac{\mathbf{b} \cdot \mathbf{a}}{\mathbf{a} \cdot \mathbf{a}} \mathbf{a} = \frac{2 + 8 + 9}{4 + 16 + 9} \mathbf{a} = \frac{19}{29} [2, 4, 3].$$

The Concept of Projection

We now explain what is meant by the *projection* of a vector $\mathbf{b}$ in $\mathbb{R}^n$ on a general subspace W of $\mathbb{R}^n$. We will show that there are unique vectors $\mathbf{b}_W$ and $\mathbf{b}_{W^\perp}$ such that

1. $\mathbf{b}_W$ is in the subspace W ;
2. $\mathbf{b}_{W^\perp}$ is orthogonal to every vector in W ; and
3. $\mathbf{b} = \mathbf{b}_W + \mathbf{b}_{W^\perp}$.

Let $W^\perp$ be the set of all vectors in $\mathbb{R}^n$ that are perpendicular to every vector in W . Properties of $W^\perp$ will appear shortly in Theorem 6.1. Figure 6.3 gives a symbolic illustration of this decomposition of $\mathbf{b}$ into a sum of a vector in W and a vector orthogonal to W . Once we have demonstrated the existence of this decomposition, we will define the *projection of $\mathbf{b}$ on W* to be the vector $\mathbf{b}_W$.

The projection $\mathbf{b}_W$ of $\mathbf{b}$ on W is the vector $\mathbf{w}$ in W that is closest to $\mathbf{b}$. That is, $\mathbf{w} = \mathbf{b}_W$ minimizes the distance $\|\mathbf{b} - \mathbf{w}\|$ from $\mathbf{b}$ to W for all $\mathbf{w}$ in W . This seems reasonable, because we have $\mathbf{b} = \mathbf{b}_W + \mathbf{b}_{W^\perp}$ and because $\mathbf{b}_{W^\perp}$ is orthogonal to every vector in W . We can demonstrate algebraically that for any $\mathbf{w} \in W$, we have $\|\mathbf{b} - \mathbf{w}\| \geq \|\mathbf{b} - \mathbf{b}_W\|$. We work with $\|\mathbf{b} - \mathbf{w}\|^2$ so we can use the dot product. Because the dot product of any vector in W and any vector in $W^\perp$ is 0, we obtain, for all $\mathbf{w} \in W$,

$$\begin{aligned} \|\mathbf{b} - \mathbf{w}\|^2 &= (\mathbf{b} - \mathbf{w}) \cdot (\mathbf{b} - \mathbf{w}) \\ &= ((\mathbf{b} - \mathbf{b}_W) + (\mathbf{b}_W - \mathbf{w})) \cdot ((\mathbf{b} - \mathbf{b}_W) + (\mathbf{b}_W - \mathbf{w})) \\ &= (\mathbf{b} - \mathbf{b}_W) \cdot (\mathbf{b} - \mathbf{b}_W) + 2 \underbrace{(\mathbf{b} - \mathbf{b}_W)}_{\text{in } W^\perp} \cdot \underbrace{(\mathbf{b}_W - \mathbf{w})}_{\text{in } W} + (\mathbf{b}_W - \mathbf{w}) \cdot (\mathbf{b}_W - \mathbf{w}) \\ &= \|\mathbf{b} - \mathbf{b}_W\|^2 + \|\mathbf{b}_W - \mathbf{w}\|^2 \geq \|\mathbf{b} - \mathbf{b}_W\|^2. \end{aligned}$$

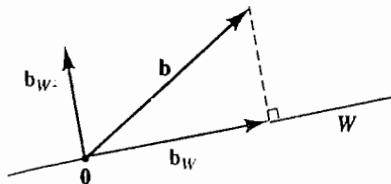


FIGURE 6.3

The decomposition $\mathbf{b} = \mathbf{b}_W + \mathbf{b}_{W^\perp}$

Of course, this minimum distance from $\mathbf{b}$ to W is just $\|\mathbf{b}_{W^\perp}\|$, as indicated by Figure 6.3.

ILLUSTRATION 2

Suppose that you are pushing a box across a floor by pushing forward and downward on the top edge. (See Figure 6.4.) We take as origin the point where the force is applied to the box and as W the plane through that origin parallel to the floor. If $\mathbf{F}$ is the force vector applied at the origin, then $\mathbf{F}_W$ is the portion of the force vector that actually moves the body along the floor, and $\mathbf{F}_{W^\perp}$ (which is directed straight down) is the portion of the force vector that attempts to push the box into the floor and thereby increases the friction between the box and the floor. ■

Orthogonal Complement of a Subspace

As a preliminary to proving the existence and uniqueness of the vectors $\mathbf{b}_W$ and $\mathbf{b}_{W^\perp}$ just described, we consider all vectors in $\mathbb{R}^n$ that are orthogonal to every vector in a subspace W .

DEFINITION 6.1 Orthogonal Complement

Let W be a subspace of $\mathbb{R}^n$. The set of all vectors in $\mathbb{R}^n$ that are orthogonal to every vector in W is the **orthogonal complement** of W , and is denoted by $W^\perp$.

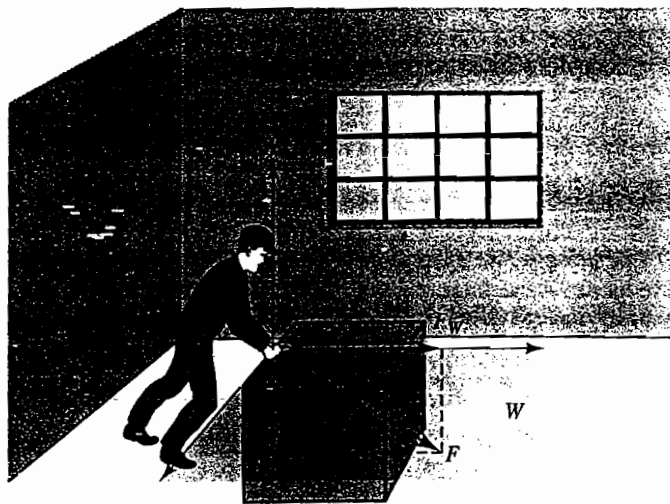


FIGURE 6.4
The decomposition of a force vector.

It is not difficult to find the orthogonal complement of W if we know a generating set for W . Let $\{v_1, v_2, \dots, v_k\}$ be a generating set for W . Let A be the $k \times n$ matrix having v_i as its i th row vector. That is,

$$A = \begin{bmatrix} \text{---} v_1 \text{---} \\ \text{---} v_2 \text{---} \\ \vdots \\ \text{---} v_k \text{---} \end{bmatrix}.$$

Thus, W is the row space of A . Now the nullspace of A consists of all vectors x in $\mathbb{R}^n$ that are solutions of the homogeneous system $Ax = 0$. But $Ax = 0$ if and only if $v_i \cdot x = 0$ for $i = 1, 2, \dots, k$. Therefore, the nullspace of A is the set of all vectors x in $\mathbb{R}^n$ that are orthogonal to each of the rows of A , and hence to the row space of A . In other words, the orthogonal complement of the row space of A is the nullspace of A . Thus we have found $W^\perp$. We summarize this procedure in a box. Notice that this procedure marks one of the rare occasions in this text when vectors must be placed as rows of a matrix.

Finding the Orthogonal Complement of a Subspace W of $\mathbb{R}^n$

1. Find a matrix A having as row vectors a generating set for W .
2. Find the nullspace of A —that is, the solution space of $Ax = 0$. This nullspace is $W^\perp$.

EXAMPLE 2 Find a basis for the orthogonal complement in $\mathbb{R}^4$ of the subspace

$$W = \text{sp}([1, 2, 2, 1], [3, 4, 2, 3]).$$

SOLUTION We find the nullspace of the matrix

$$A = \begin{bmatrix} 1 & 2 & 2 & 1 \\ 3 & 4 & 2 & 3 \end{bmatrix}.$$

Reducing A , we have

$$\begin{bmatrix} 1 & 2 & 2 & 1 \\ 3 & 4 & 2 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 2 & 1 \\ 0 & -2 & -4 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -2 & 1 \\ 0 & 1 & 2 & 0 \end{bmatrix}.$$

Therefore, the nullspace of A , which is the orthogonal complement of W , is the set of vectors of the form

$$[2r - s, -2r, r, s] \quad \text{for any scalars } r \text{ and } s.$$

Thus $\{[2, -2, 1, 0], [-1, 0, 0, 1]\}$ is a basis for $W^\perp$. ■

ILLUSTRATION 3 Note that if W is the $(n-1)$ -dimensional solution space in $\mathbb{R}^n$ of a single linear equation $a_1x_1 + a_2x_2 + \cdots + a_nx_n = 0$, then the vector $[a_1, a_2, \dots, a_n]$ of coefficients is orthogonal to W , because the equation can be written as

$$[a_1, a_2, \dots, a_n] \cdot [x_1, x_2, \dots, x_n] = 0.$$

Thus $W^\perp = \text{sp}([a_1, a_2, \dots, a_n])$. The subspace W is a line if $n = 2$, a plane if $n = 3$, and is called a *hyperplane* for other values of n . ■

We now show that the orthogonal complement of a subspace W has some very nice properties. In particular, we exhibit the decomposition $\mathbf{b} = \mathbf{b}_W + \mathbf{b}_{W^\perp}$ described earlier and show that it is unique.

THEOREM 6.1 Properties of $W^\perp$

The orthogonal complement $W^\perp$ of a subspace W of $\mathbb{R}^n$ has the following properties:

1. $W^\perp$ is a subspace of $\mathbb{R}^n$.
2. $\dim(W^\perp) = n - \dim(W)$.
3. $(W^\perp)^\perp = W$; that is, the orthogonal complement of $W^\perp$ is W .
4. Each vector $\mathbf{b}$ in $\mathbb{R}^n$ can be expressed uniquely in the form $\mathbf{b} = \mathbf{b}_W + \mathbf{b}_{W^\perp}$ for $\mathbf{b}_W$ in W and $\mathbf{b}_{W^\perp}$ in $W^\perp$.

PROOF We may assume $W \neq \{0\}$. Let $\dim(W) = k$, and let $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ be a basis for W . Let A be the $k \times n$ matrix having $\mathbf{v}_i$ as its i th row vector for $i = 1, \dots, k$.

For property 1, we have seen that $W^\perp$ is the nullspace of the matrix A , so it is a subspace of $\mathbb{R}^n$.

For property 2, consider the rank equation of A :

$$\text{rank}(A) + \text{nullity}(A) = n.$$

Because $\dim(W) = \text{rank}(A)$ and because $W^\perp$ is the nullspace of A , we see that $\dim(W^\perp) = n - \dim(W)$.

For property 3, applying property 2 with the subspace $W^\perp$, we find that

$$\dim(W^\perp)^\perp = n - \dim(W^\perp) = n - (n - k) = k.$$

However, every vector in W is orthogonal to every vector in $W^\perp$, so W is a subspace of $(W^\perp)^\perp$. Because both W and $(W^\perp)^\perp$ have the same dimension k , we conclude that W must be equal to $(W^\perp)^\perp$. (See Exercise 38 in Section 2.1.)

For property 4, let $\{v_{k+1}, v_{k+2}, \dots, v_n\}$ be a basis for $W^\perp$. We claim that the set

$$\{v_1, v_2, \dots, v_n\} \quad (2)$$

is a basis for $\mathbb{R}^n$. Consider a relation

$$r_1 v_1 + r_2 v_2 + \dots + r_k v_k + s_{k+1} v_{k+1} + s_{k+2} v_{k+2} + \dots + s_n v_n = 0.$$

Rewrite this relation as

$$r_1 v_1 + r_2 v_2 + \dots + r_k v_k = -s_{k+1} v_{k+1} - s_{k+2} v_{k+2} - \dots - s_n v_n. \quad (3)$$

The sum on the left-hand side is in W , and the sum on the right-hand side is in $W^\perp$. Because these sums are equal, they represent a vector that is in both W and $W^\perp$ and must be orthogonal to itself. The only such vector is the zero vector, so both sides of Eq. (3) must equal 0. Because the v_i are independent for $1 \leq i \leq k$ and the v_j are independent for $k+1 \leq j \leq n$, we see that all the scalars r_i and s_j are zero. This shows that set (2) is independent; because it contains n vectors, it must be a basis for $\mathbb{R}^n$. Therefore, for every vector b in $\mathbb{R}^n$ we can express b in the form

$$b = \underbrace{r_1 v_1 + r_2 v_2 + \dots + r_k v_k}_{b_W} + \underbrace{s_{k+1} v_{k+1} + s_{k+2} v_{k+2} + \dots + s_n v_n}_{b_{W^\perp}}.$$

This shows that b can indeed be expressed as a sum of a vector in W and a vector in $W^\perp$, and this expression is unique because each vector in $\mathbb{R}^n$ is a unique combination of the vectors in the basis $\{v_1, v_2, \dots, v_n\}$. $\blacktriangle$

Projection of a Vector on a Subspace

Now that the groundwork is firmly established, we can define the projection of a vector in $\mathbb{R}^n$ on a subspace and then illustrate with some examples.

DEFINITION 6.2 Projection of b on W

Let b be a vector in $\mathbb{R}^n$, and let W be a subspace of $\mathbb{R}^n$. Let

$$b = b_W + b_{W^\perp},$$

as described in Theorem 6.1. Then b_W is the projection of b on W .

Theorem 6.1 shows that the projection of b on W is unique. It also shows one way in which it can be computed.

Steps to Find the Projection of $\mathbf{b}$ on W

1. Select a basis $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ for the subspace W . (Often this is given.)
2. Find a basis $\{\mathbf{v}_{k+1}, \mathbf{v}_{k+2}, \dots, \mathbf{v}_n\}$ for $W^\perp$, as in Example 2.
3. Find the coordinate vector $\mathbf{r} = [r_1, r_2, \dots, r_n]$ of $\mathbf{b}$ relative to the basis $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ so that $\mathbf{b} = r_1\mathbf{v}_1 + r_2\mathbf{v}_2 + \dots + r_n\mathbf{v}_n$. (See the box on page 207.)
4. Then $\mathbf{b}_W = r_1\mathbf{v}_1 + r_2\mathbf{v}_2 + \dots + r_k\mathbf{v}_k$.

EXAMPLE 3 Find the projection of $\mathbf{b} = [2, 1, 5]$ on the subspace $W = \text{sp}([1, 2, 1], [2, 1, -1])$.

SOLUTION We follow the boxed procedure.

Step 1: Because $\mathbf{v}_1 = [1, 2, 1]$ and $\mathbf{v}_2 = [2, 1, -1]$ are independent, they form a basis for W .

Step 2: A basis for $W^\perp$ can be found by obtaining the nullspace of the matrix

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & -1 \end{bmatrix}.$$

An echelon form of A is

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & -3 & -3 \end{bmatrix},$$

and the nullspace of A is the set of vectors $[r, -r, r]$, where r is any scalar. Let us take $\mathbf{v}_3 = [1, -1, 1]$ to form the basis $\{\mathbf{v}_3\}$ of $W^\perp$. (Alternatively, we could have computed $\mathbf{v}_1 \times \mathbf{v}_2$ to find a suitable $\mathbf{v}_3$.)

Step 3: To find the coordinate vector $\mathbf{r}$ of $\mathbf{b}$ relative to the ordered basis $(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$, we proceed as described in Section 3.3, and perform the reduction

$$\begin{array}{ccc|c} \begin{bmatrix} 1 & 2 & 1 & 2 \\ 2 & 1 & -1 & 1 \\ 1 & -1 & 1 & 5 \end{bmatrix} & \sim & \begin{bmatrix} 1 & 2 & 1 & 2 \\ 0 & -3 & -3 & -3 \\ 0 & -3 & 0 & 3 \end{bmatrix} & \sim & \begin{bmatrix} 1 & 0 & 1 & 4 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & -3 & -6 \end{bmatrix} \\ \mathbf{v}_1 & \mathbf{v}_2 & \mathbf{v}_3 & \mathbf{b} & & & & \\ & & & & \sim & \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 2 \end{bmatrix} & & \end{array}$$

Thus, $\mathbf{r} = [2, -1, 2]$.

Step 4: The projection of $\mathbf{b}$ on W is

$$\mathbf{b}_W = 2\mathbf{v}_1 - \mathbf{v}_2 = 2[1, 2, 1] - [2, 1, -1] = [0, 3, 3].$$

As a check, notice that $2\mathbf{v}_3 = [2, -2, 2]$ is the projection of $\mathbf{b}$ on $W^\perp$, and $\mathbf{b} = \mathbf{b}_W + \mathbf{b}_{W^\perp} = [0, 3, 3] + [2, -2, 2] = [2, 1, 5]$. ■

Our next example shows that the procedure described in the box preceding Example 3 yields formula (1) for the projection of a vector $\mathbf{b}$ on $\text{sp}(\mathbf{a})$.

EXAMPLE 4 Let $\mathbf{a} \neq \mathbf{0}$ and $\mathbf{b}$ be vectors in $\mathbb{R}^n$. Find the projection of $\mathbf{b}$ on $\text{sp}(\mathbf{a})$, using the same boxed procedure we have been applying.

SOLUTION We project $\mathbf{b}$ on the subspace $W = \text{sp}(\mathbf{a})$ of $\mathbb{R}^n$. Let $\{\mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_n\}$ be a basis for $W^\perp$, and let $\mathbf{r} = [r_1, r_2, \dots, r_n]$ be the coordinate vector of $\mathbf{b}$ relative to the ordered basis $(\mathbf{a}, \mathbf{v}_2, \dots, \mathbf{v}_n)$. Then

$$\mathbf{b} = r_1 \mathbf{a} + r_2 \mathbf{v}_2 + \dots + r_n \mathbf{v}_n.$$

Because $\mathbf{v}_i \cdot \mathbf{a} = 0$ for $i = 2, \dots, n$, we see that $\mathbf{b} \cdot \mathbf{a} = r_1 \mathbf{a} \cdot \mathbf{a} = r_1(\mathbf{a} \cdot \mathbf{a})$. Because $\mathbf{a} \neq \mathbf{0}$, we can write

$$r_1 = \frac{\mathbf{b} \cdot \mathbf{a}}{\mathbf{a} \cdot \mathbf{a}}.$$

The projection of $\mathbf{b}$ on $\text{sp}(\mathbf{a})$ is then $r_1 \mathbf{a} = \frac{\mathbf{b} \cdot \mathbf{a}}{\mathbf{a} \cdot \mathbf{a}} \mathbf{a}$.

We have seen that if W is a one-dimensional subspace of $\mathbb{R}^n$, then we can readily find the projection $\mathbf{b}_W$ of $\mathbf{b} \in \mathbb{R}^n$ on W . We simply use formula (1). On the other hand, if $W^\perp$ is one-dimensional, we can use formula (1) to find the projection of $\mathbf{b}$ on $W^\perp$, and then find $\mathbf{b}_W$ using the relation $\mathbf{b}_W = \mathbf{b} - \mathbf{b}_{W^\perp}$. Our next example illustrates this.

EXAMPLE 5 Find the projection of the vector $[3, -1, 2]$ on the plane $x + y + z = 0$ through the origin in $\mathbb{R}^3$.

SOLUTION Let W be the subspace of $\mathbb{R}^3$ given by the plane $x + y + z = 0$. Then $W^\perp$ is one-dimensional, and a generating vector for $W^\perp$ is $\mathbf{a} = [1, 1, 1]$, obtained by taking the coefficients of x, y , and z in this equation. (See Illustration 3.) Let $\mathbf{b} = [3, -1, 2]$. By formula (1), we have

$$\mathbf{b}_{W^\perp} = \frac{\mathbf{b} \cdot \mathbf{a}}{\mathbf{a} \cdot \mathbf{a}} \mathbf{a} = \frac{3 - 1 + 2}{1 + 1 + 1} [1, 1, 1] = \frac{4}{3} [1, 1, 1].$$

Thus $\mathbf{b}_W = \mathbf{b} - \mathbf{b}_{W^\perp} = [3, -1, 2] - \frac{4}{3} [1, 1, 1] = \left[\frac{5}{3}, -\frac{7}{3}, \frac{2}{3}\right]$.

Projections in Inner-Product Spaces (Optional)

Everything we have done in this section is equally valid for any finite-dimensional inner-product space. For example, formula (1) for projecting one vector on a one-dimensional subspace takes the form

$$\mathbf{p} = \frac{\langle \mathbf{b}, \mathbf{a} \rangle}{\langle \mathbf{a}, \mathbf{a} \rangle} \mathbf{a}. \quad (4)$$

The fact that the projection $\mathbf{b}_W$ of a vector $\mathbf{b}$ on a subspace W is the vector $\mathbf{w}$ in W that minimizes $\|\mathbf{b} - \mathbf{w}\|$ is useful in many contexts.

For an example involving function spaces, suppose that f is a complicated function and that W consists of functions that are easily handled, such as

polynomials or trigonometric functions. Using a suitable inner product and projection, we find that the function f_W in W becomes a best approximation to the function f by functions in W . Example 6 illustrates this by approximating the function $f(x) = x$ over the interval $0 \leq x \leq 1$ by a function in the space W of all constant functions on this interval. Visualizing their graphs, we are not surprised that the constant function $p(x) = \frac{1}{2}$ turns out to be the best approximation to x in W , using the inner product we defined on function spaces in Section 3.5. In Section 6.5, we will use this minimization feature again to find a best approximate solution of an inconsistent overdetermined linear system.

EXAMPLE 6 Let the inner product of two polynomials $p(x)$ and $q(x)$ in the space $P_{0,1}$ of polynomial functions with domain $0 \leq x \leq 1$ be defined by

$$\langle p(x), q(x) \rangle = \int_0^1 p(x)q(x) dx.$$

(See Example 3 in Section 3.5.) Find the projection of $f(x) = x$ on $\text{sp}(1)$, using formula (4). Then find the projection of x on $\text{sp}(1)^\perp$.

SOLUTION By formula (4), the projection of x on $\text{sp}(1)$ is

$$\left(\frac{\langle x, 1 \rangle}{\langle 1, 1 \rangle} \right) 1 = \frac{\int_0^1 (x)(1) dx}{\int_0^1 (1)(1) dx} 1 = \left(\frac{1/2}{1} \right) 1 = \frac{1}{2}.$$

The projection of x on $\text{sp}(1)^\perp$ is obtained by subtracting the projection on $\text{sp}(1)$ from x . We obtain $x - \frac{1}{2}$. As a check, we should then have $\langle \frac{1}{2}, x - \frac{1}{2} \rangle = 0$, and a computation of $\int_0^1 (\frac{1}{2})(x - \frac{1}{2}) dx$ shows that this is indeed so. ■

SUMMARY

1. The projection of $\mathbf{b}$ in $\mathbb{R}^n$ on $\text{sp}(\mathbf{a})$ for a nonzero vector $\mathbf{a}$ in $\mathbb{R}^n$ is given by $((\mathbf{b} \cdot \mathbf{a})/(\mathbf{a} \cdot \mathbf{a}))\mathbf{a}$.
2. The orthogonal complement $W^\perp$ of a subspace W of $\mathbb{R}^n$ is the set of all vectors in $\mathbb{R}^n$ that are orthogonal to every vector in W . Further, $W^\perp$ is a subspace of $\mathbb{R}^n$ of dimension $n - \dim(W)$, and $(W^\perp)^\perp = W$.
3. The row space and the nullspace of an $m \times n$ matrix A are orthogonal complements of each other. In particular, $W^\perp$ can be computed as the nullspace of a matrix A having as its row vectors the vectors in a generating set for W .
4. Let W be a subspace of $\mathbb{R}^n$. Each vector $\mathbf{b}$ in $\mathbb{R}^n$ can be expressed uniquely in the form $\mathbf{b} = \mathbf{b}_W + \mathbf{b}_{W^\perp}$ for $\mathbf{b}_W$ in W and $\mathbf{b}_{W^\perp}$ in $W^\perp$.
5. The vectors $\mathbf{b}_W$ and $\mathbf{b}_{W^\perp}$ are the projections of $\mathbf{b}$ on W and on $W^\perp$, respectively. They can be computed by means of the boxed procedure on page 333.

EXERCISES

In Exercises 1–6, find the indicated projection.

1. The projection of $[2, 1]$ on $\text{sp}([3, 4])$ in $\mathbb{R}^2$
2. The projection of $[3, 4]$ on $\text{sp}([2, 1])$ in $\mathbb{R}^2$
3. The projection of $[1, 2, 1]$ on each of the unit coordinate vectors in $\mathbb{R}^3$
4. The projection of $[1, 2, 1]$ on the line with parametric equations $x = 3t, y = t, z = 2t$ in $\mathbb{R}^3$
5. The projection of $[-1, 2, 0, 1]$ on $\text{sp}([2, -3, 1, 2])$ in $\mathbb{R}^4$
6. The projection of $[2, -1, 3, -5]$ on the line $\text{sp}([1, 0, -1, 2])$ in $\mathbb{R}^4$

In Exercises 7–12, find the orthogonal complement of the given subspace.

7. The subspace $\text{sp}([1, 2, -1])$ in $\mathbb{R}^3$
8. The line $\text{sp}([2, -1, 0, -3])$ in $\mathbb{R}^4$
9. The subspace $\text{sp}([1, 3, 0], [2, 1, 4])$ in $\mathbb{R}^3$
10. The plane $2x + y + 3z = 0$ in $\mathbb{R}^3$
11. The subspace $\text{sp}([2, 1, 3, 4], [1, 0, -2, 1])$ in $\mathbb{R}^4$
12. The subspace (hyperplane) $ax_1 + bx_2 + cx_3 + dx_4 = 0$ in $\mathbb{R}^4$ [HINT: See Illustration 3.]
13. Find a nonzero vector in $\mathbb{R}^3$ perpendicular to $[1, 1, 2]$ and $[2, 3, 1]$ by
 - a. the methods of this section,
 - b. computing a determinant.
14. Find a nonzero vector in $\mathbb{R}^4$ perpendicular to $[1, 0, -1, 1]$, $[0, 0, -1, 1]$, and $[2, -1, 2, 0]$ by
 - a. the methods of this section,
 - b. computing a determinant.

In Exercises 15–22, find the indicated projection.

15. The projection of $[1, 2, 1]$ on the subspace $\text{sp}([3, 1, 2], [1, 0, 1])$ in $\mathbb{R}^3$
16. The projection of $[1, 2, 1]$ on the plane $x + y + z = 0$ in $\mathbb{R}^3$
17. The projection of $[1, 0, 0]$ on the subspace $\text{sp}([2, 1, 1], [1, 0, 2])$ in $\mathbb{R}^3$
18. The projection of $[-1, 0, 1]$ on the plane $x + y = 0$ in $\mathbb{R}^3$
19. The projection of $[0, 0, 1]$ on the plane $2x - y - z = 0$ in $\mathbb{R}^3$
20. The projection in $\mathbb{R}^4$ of $[-2, 1, 3, -5]$ on
 - a. the subspace $\text{sp}(e_3)$
 - b. the subspace $\text{sp}(e_1, e_4)$
 - c. the subspace $\text{sp}(e_1, e_3, e_4)$
 - d. $\mathbb{R}^4$
21. The projection of $[1, 0, -1, 1]$ on the subspace $\text{sp}([1, 0, 0, 0], [0, 1, 1, 0], [0, 0, 1, 1])$ in $\mathbb{R}^4$
22. The projection of $[0, 1, -1, 0]$ on the subspace (hyperplane) $x_1 - x_2 + x_3 + x_4 = 0$ in $\mathbb{R}^4$ [HINT: See Example 5.]
23. Assume that $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ are vectors in $\mathbb{R}^n$ and that W is a subspace of $\mathbb{R}^n$. Mark each of the following True or False.
 - a. The projection of $\mathbf{b}$ on $\text{sp}(\mathbf{a})$ is a scalar multiple of $\mathbf{b}$.
 - b. The projection of $\mathbf{b}$ on $\text{sp}(\mathbf{a})$ is a scalar multiple of $\mathbf{a}$.
 - c. The set of all vectors in $\mathbb{R}^n$ orthogonal to every vector in W is a subspace of $\mathbb{R}^n$.
 - d. The vector $\mathbf{w} \in W$ that minimizes $\|\mathbf{c} - \mathbf{w}\|$ is $\mathbf{c}_W$.
 - e. If the projection of $\mathbf{b}$ on W is $\mathbf{b}$ itself, then $\mathbf{b}$ is orthogonal to every vector in W .
 - f. If the projection of $\mathbf{b}$ on W is $\mathbf{b}$ itself, then $\mathbf{b}$ is in W .
 - g. The vector $\mathbf{b}$ is orthogonal to every vector in W if and only if $\mathbf{b}_W = \mathbf{0}$.
 - h. The intersection of W and $W^\perp$ is empty.
 - i. If $\mathbf{b}$ and $\mathbf{c}$ have the same projection on W , then $\mathbf{b} = \mathbf{c}$.
 - j. If $\mathbf{b}$ and $\mathbf{c}$ have the same projection on every subspace of $\mathbb{R}^n$, then $\mathbf{b} = \mathbf{c}$.
24. Let $\mathbf{a}$ and $\mathbf{b}$ be nonzero vectors in $\mathbb{R}^n$, and let θ be the angle between $\mathbf{a}$ and $\mathbf{b}$. The scalar $\|\mathbf{b}\| \cos \theta$ is called the **scalar component of $\mathbf{b}$ along $\mathbf{a}$** . Interpret this scalar graphically (see Figures 6.1 and 6.2), and give a formula for it in terms of the dot product.
25. Let W be a subspace of $\mathbb{R}^n$ and let $\mathbf{b}$ be a vector in $\mathbb{R}^n$. Prove that there is one and only one vector $\mathbf{p}$ in W such that $\mathbf{b} - \mathbf{p}$ is

perpendicular to every vector in W . [HINT: Suppose that $\mathbf{p}_1$ and $\mathbf{p}_2$ are two such vectors, and show that $\mathbf{p}_1 - \mathbf{p}_2$ is in $W^\perp$.]

26. Let A be an $m \times n$ matrix.
- Prove that the set W of row vectors $\mathbf{x}$ in $\mathbb{R}^m$ such that $\mathbf{x}A = \mathbf{0}$ is a subspace of $\mathbb{R}^m$.
 - Prove that the subspace W in part (a) and the column space of A are orthogonal complements.
27. Subspaces U and W of $\mathbb{R}^n$ are orthogonal if $\mathbf{u} \cdot \mathbf{w} = 0$ for all $\mathbf{u}$ in U and all $\mathbf{w}$ in W . Let U and W be orthogonal subspaces of $\mathbb{R}^n$, and let $\dim(U) = n - \dim(W)$. Prove that each subspace is the orthogonal complement of the other.
28. Let W be a subspace of $\mathbb{R}^n$ with orthogonal complement $W^\perp$. Writing $\mathbf{a} = \mathbf{a}_W + \mathbf{a}_{W^\perp}$, as in Theorem 6.1, prove that

$$\|\mathbf{a}\| = \sqrt{\|\mathbf{a}_W\|^2 + \|\mathbf{a}_{W^\perp}\|^2}.$$

[HINT: Use the formula $\|\mathbf{a}\|^2 = \mathbf{a} \cdot \mathbf{a}$.]

29. (Distance from a point to a subspace) Let W be a subspace of $\mathbb{R}^n$. Figure 6.5 suggests that the distance from the tip of $\mathbf{a}$ in $\mathbb{R}^n$ to the subspace W is equal to the magnitude of the projection of the vector $\mathbf{a}$ on the orthogonal complement of W . Find the distance from the point $(1, 2, 3)$ in $\mathbb{R}^3$ to the subspace (plane) $\text{sp}\{[2, 2, 1], [1, 2, 1]\}$.

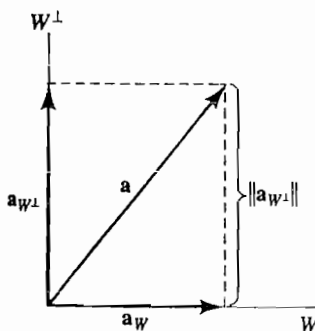


FIGURE 6.5

The distance from $\mathbf{a}$ to W is $\|\mathbf{a}_{W^\perp}\|$.

30. Find the distance from the point $(2, 1, 3, 1)$ in $\mathbb{R}^4$ to the plane $\text{sp}\{[1, 0, 1, 0], [1, -1, 1, 1]\}$. [HINT: See Exercise 29.]

In Exercises 31–36, use the idea in Exercise 29 to find the distance from the tip of $\mathbf{a}$ to the given one-dimensional subspace (line). [NOTE: To calculate $\|\mathbf{a}_{W^\perp}\|$, first calculate $\|\mathbf{a}_W\|$ and then use Exercise 28.]

- $\mathbf{a} = [1, 2, 1]$,
 $W = \text{sp}\{[2, 1, 0]\}$ in $\mathbb{R}^3$
- $\mathbf{a} = [2, -1, 3]$,
 $W = \text{sp}\{[1, 2, 4]\}$ in $\mathbb{R}^3$
- $\mathbf{a} = [1, 2, -1, 0]$,
 $W = \text{sp}\{[3, 1, 4, -1]\}$ in $\mathbb{R}^4$
- $\mathbf{a} = [2, 1, 1, 2]$,
 $W = \text{sp}\{[1, 2, 1, 3]\}$ in $\mathbb{R}^4$
- $\mathbf{a} = [1, 2, 3, 4, 5]$,
 $W = \text{sp}\{[1, 1, 1, 1, 1]\}$ in $\mathbb{R}^5$
- $\mathbf{a} = [1, 0, 1, 0, 1, 0, 1]$,
 $W = \text{sp}\{[1, 2, 3, 4, 3, 2, 1]\}$ in $\mathbb{R}^7$

Exercises 37–39 involve inner-product spaces discussed in optional Section 3.5.

- Referring to Example 6, find the projection of $f(x) = 1$ on $\text{sp}(x)$ in P_2 .
- Referring to Example 6, find the projection of $f(x) = x$ on $\text{sp}(1 + x)$.
- Let S and T be nonempty subsets of an inner-product space V with the property that every vector in S is orthogonal to every vector in T . Prove that the span of S and the span of T are orthogonal subspaces of V .
- Work with Topic 3 of the routine VECTGRPH in LINTEK until you are able to get a score of at least 80% most of the time.

6.2

THE GRAM-SCHMIDT PROCESS

In the preceding section, we saw how to project a vector $\mathbf{b}$ on a subspace W of $\mathbb{R}^n$. The calculations can be somewhat tedious. We open this section by observing that, if we know a basis for W consisting of mutually perpendicular vectors, the computational burden can be eased. We then present the Gram-Schmidt algorithm, showing how such a nice basis for W can be found.

Orthogonal and Orthonormal Bases

A set $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ of nonzero vectors in $\mathbb{R}^n$ is **orthogonal** if the vectors $\mathbf{v}_i$ are mutually perpendicular—that is, if $\mathbf{v}_i \cdot \mathbf{v}_j = 0$ for $i \neq j$. Our next theorem shows that an orthogonal generating set for a subspace of $\mathbb{R}^n$ is sure to be a basis.

THEOREM 6.2 Orthogonal Bases

Let $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ be an orthogonal set of nonzero vectors in $\mathbb{R}^n$. Then this set is independent and consequently is a basis for the subspace $\text{sp}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k)$.

PROOF To show that the orthogonal set $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ is independent, let us suppose that

$$\mathbf{v}_j = s_1 \mathbf{v}_1 + s_2 \mathbf{v}_2 + \dots + s_{j-1} \mathbf{v}_{j-1}$$

Taking the dot product of both sides of this equation with $\mathbf{v}_j$ yields $\mathbf{v}_j \cdot \mathbf{v}_j = 0$, which contradicts the hypothesis that $\mathbf{v}_j \neq \mathbf{0}$. Thus, no $\mathbf{v}_j$ is a linear combination of its predecessors, so $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ is independent and thus is a basis for $\text{sp}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k)$. (See Exercise 37 on page 203.) $\blacktriangle$

EXAMPLE 1 Find an orthogonal basis for the plane $2x - y + z = 0$ in $\mathbb{R}^3$.

SOLUTION The given plane contains the origin and hence is a subspace of $\mathbb{R}^3$. We need only find two perpendicular vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ in this plane. Letting $y = 0$ and $z = 2$, we find that $x = -1$ in the given equation, so $\mathbf{v}_1 = [-1, 0, 2]$ lies in the plane. Because the vector $[2, -1, 1]$ of coefficients is perpendicular to the plane, we compute a cross product, and let

$$\mathbf{v}_2 = [-1, 0, 2] \times [2, -1, 1] = [2, 5, 1].$$

This vector is perpendicular to the coefficient vector $[2, -1, 1]$, so it lies in the plane; and of course, it is also perpendicular to the vector $[-1, 0, 2]$. Thus, $\{\mathbf{v}_1, \mathbf{v}_2\}$ is an orthogonal basis for the plane. $\blacksquare$

Now we show how easy it is to project a vector $\mathbf{b}$ on a subspace W of $\mathbb{R}^n$ if we know an orthogonal basis $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ for W . Recall from Section 6.1 that

$$\mathbf{b} = \mathbf{b}_W + \mathbf{b}_{W^\perp}, \quad (1)$$

where $\mathbf{b}_W$ is the projection of $\mathbf{b}$ on W , and $\mathbf{b}_{W^\perp}$ is the projection of $\mathbf{b}$ on $W^\perp$. Because $\mathbf{b}_W$ lies in W , we have

$$\mathbf{b}_W = r_1 \mathbf{v}_1 + r_2 \mathbf{v}_2 + \dots + r_k \mathbf{v}_k \quad (2)$$

for some choice of scalars r_i . Computing the dot product of $\mathbf{b}$ with $\mathbf{v}_i$ and using Eqs. (1) and (2), we have

$$\begin{aligned} \mathbf{b} \cdot \mathbf{v}_i &= (\mathbf{b}_W \cdot \mathbf{v}_i) + (\mathbf{b}_{W^\perp} \cdot \mathbf{v}_i) \\ &= (r_1 \mathbf{v}_1 \cdot \mathbf{v}_i + r_2 \mathbf{v}_2 \cdot \mathbf{v}_i + \dots + r_k \mathbf{v}_k \cdot \mathbf{v}_i) + 0 \quad \mathbf{v}_i \text{ is in } W \\ &= r_i \mathbf{v}_i \cdot \mathbf{v}_i \quad \mathbf{v}_i \cdot \mathbf{v}_j = 0 \text{ for } i \neq j \end{aligned}$$

Therefore, $r_i = (\mathbf{b} \cdot \mathbf{v}_i) / (\mathbf{v}_i \cdot \mathbf{v}_i)$, so

$$r_i \mathbf{v}_i = \frac{\mathbf{b} \cdot \mathbf{v}_i}{\mathbf{v}_i \cdot \mathbf{v}_i} \mathbf{v}_i,$$

which is just the projection of $\mathbf{b}$ on $\mathbf{v}_i$. In other words, to project $\mathbf{b}$ on W , we need only project $\mathbf{b}$ on each of the orthogonal basis vectors, and then add! We summarize this in a theorem.

THEOREM 6.3 Projection Using an Orthogonal Basis

Let $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ be an orthogonal basis for a subspace W of $\mathbb{R}^n$, and let $\mathbf{b}$ be any vector in $\mathbb{R}^n$. The projection of $\mathbf{b}$ on W is

$$\mathbf{b}_W = \frac{\mathbf{b} \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1} \mathbf{v}_1 + \frac{\mathbf{b} \cdot \mathbf{v}_2}{\mathbf{v}_2 \cdot \mathbf{v}_2} \mathbf{v}_2 + \dots + \frac{\mathbf{b} \cdot \mathbf{v}_k}{\mathbf{v}_k \cdot \mathbf{v}_k} \mathbf{v}_k. \quad (3)$$

EXAMPLE 2 Find the projection of $\mathbf{b} = [3, -2, 2]$ on the plane $2x - y + z = 0$ in $\mathbb{R}^3$.

SOLUTION In Example 1, we found an orthogonal basis for the given plane, consisting of the vectors $\mathbf{v}_1 = [-1, 0, 2]$ and $\mathbf{v}_2 = [2, 5, 1]$. Thus, the plane may be expressed as $W = \text{sp}(\mathbf{v}_1, \mathbf{v}_2)$. Using Eq. (3), we have

$$\begin{aligned} \mathbf{b}_W &= \frac{\mathbf{b} \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1} \mathbf{v}_1 + \frac{\mathbf{b} \cdot \mathbf{v}_2}{\mathbf{v}_2 \cdot \mathbf{v}_2} \mathbf{v}_2 \\ &= \frac{1}{5}[-1, 0, 2] + \left(-\frac{2}{30}\right)[2, 5, 1] = \frac{3}{15}[-1, 0, 2] - \frac{1}{15}[2, 5, 1] \\ &= \left[-\frac{1}{3}, -\frac{1}{3}, \frac{1}{3}\right]. \end{aligned}$$

It is sometimes desirable to *normalize* the vectors in an orthogonal basis, converting each basis vector to one parallel to it but of unit length. The result remains a basis for the same subspace. Notice that the standard basis in $\mathbb{R}^n$ consists of such *perpendicular unit vectors*. Such bases are extremely useful and merit a formal definition.

DEFINITION 6.3 Orthonormal Basis

Let W be a subspace of $\mathbb{R}^n$. A basis $\{q_1, q_2, \dots, q_k\}$ for W is **orthonormal** if

1. $q_i \cdot q_j = 0$ for $i \neq j$, **Mutually perpendicular**
2. $q_i \cdot q_i = 1$. **Length 1**

The standard basis for $\mathbb{R}^n$ is just one of many orthonormal bases for $\mathbb{R}^n$ if $n > 1$. For example, any two perpendicular vectors v_1 and v_2 on the unit circle (illustrated in Figure 6.6) form an orthonormal basis for $\mathbb{R}^2$.

For the projection of a vector on a subspace that has a known orthonormal basis, Eq. (3) in Theorem 6.3 assumes a simpler form:

Projection of b on W with Orthonormal Basis $\{q_1, q_2, \dots, q_k\}$

$$b_W = (b \cdot q_1)q_1 + (b \cdot q_2)q_2 + \cdots + (b \cdot q_k)q_k \quad (4)$$

EXAMPLE 3 Find an orthonormal basis for $W = \text{sp}(v_1, v_2, v_3)$ in $\mathbb{R}^4$ if $v_1 = [1, 1, 1, 1]$, $v_2 = [-1, 1, -1, 1]$, and $v_3 = [1, -1, -1, 1]$. Then find the projection of $b = [1, 2, 3, 4]$ on W .

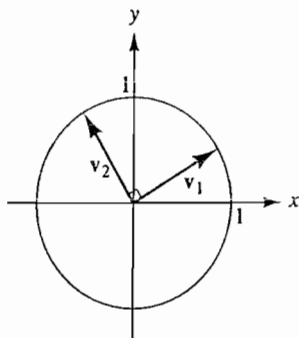


FIGURE 6.6
One of many orthonormal bases for $\mathbb{R}_2$.

SOLUTION We see that $\mathbf{v}_1 \cdot \mathbf{v}_2 = \mathbf{v}_2 \cdot \mathbf{v}_3 = \mathbf{v}_1 \cdot \mathbf{v}_3 = 0$. Because $\|\mathbf{v}_1\| = \|\mathbf{v}_2\| = \|\mathbf{v}_3\| = 2$, let $\mathbf{q}_i = \frac{1}{2}\mathbf{v}_i$ for $i = 1, 2, 3$, to obtain an orthonormal basis $\{\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3\}$ for W , so that

$$\mathbf{q}_1 = \left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2} \right], \quad \mathbf{q}_2 = \left[-\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}, \frac{1}{2} \right], \quad \text{and} \quad \mathbf{q}_3 = \left[\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}, \frac{1}{2} \right].$$

To find the projection of $\mathbf{b} = [1, 2, 3, 4]$ on W , we use Eq. (4) and obtain

$$\begin{aligned} \mathbf{b}_W &= (\mathbf{b} \cdot \mathbf{q}_1)\mathbf{q}_1 + (\mathbf{b} \cdot \mathbf{q}_2)\mathbf{q}_2 + (\mathbf{b} \cdot \mathbf{q}_3)\mathbf{q}_3 \\ &= 5\mathbf{q}_1 + \mathbf{q}_2 + 0\mathbf{q}_3 = [2, 3, 2, 3]. \end{aligned}$$

The Gram-Schmidt Process

We now describe a computational technique for creating an orthonormal basis from a given basis of a subspace W of $\mathbb{R}^n$. The theorem that follows asserts the existence of such a basis; its proof is constructive. That is, the proof shows how an orthonormal basis can be constructed.

THEOREM 6.4 Orthonormal Basis (Gram-Schmidt) Theorem

Let W be a subspace of $\mathbb{R}^n$, let $\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_k\}$ be any basis for W , and let

$$W_j = \text{sp}(\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_j) \quad \text{for } j = 1, 2, \dots, k.$$

Then there is an orthonormal basis $\{\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_k\}$ for W such that $W_j = \text{sp}(\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_j)$.

PROOF Let $\mathbf{v}_1 = \mathbf{a}_1$. For $j = 2, \dots, k$, let $\mathbf{p}_j$ be the projection of $\mathbf{a}_j$ on W_{j-1} , and let $\mathbf{v}_j = \mathbf{a}_j - \mathbf{p}_j$. That is, $\mathbf{v}_j$ is obtained by subtracting from $\mathbf{a}_j$ its projection on the subspace generated by its predecessors. Figure 6.7 gives a symbolic illustration. The decomposition

$$\mathbf{a}_j = \mathbf{p}_j + (\mathbf{a}_j - \mathbf{p}_j)$$

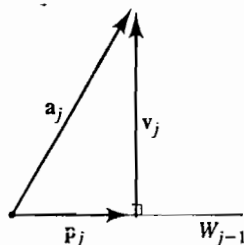


FIGURE 6.7
The vector $\mathbf{v}_j$ in the Gram-Schmidt construction.

is the unique expression for $\mathbf{a}_j$ as the sum of the vector $\mathbf{p}_j$ in W_{j-1} and the vector $\mathbf{a}_j - \mathbf{p}_j$ in $(W_{j-1})^\perp$, described in Theorem 6.1. Because $\mathbf{a}_j$ is in W_j and because $\mathbf{p}_j$ is in W_{j-1} , which is itself contained in W_j , we see that $\mathbf{v}_j = \mathbf{a}_j - \mathbf{p}_j$ lies in the subspace W_j . Now $\mathbf{v}_j$ is perpendicular to every vector in W_{j-1} . Consequently, $\mathbf{v}_j$ is perpendicular to $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{j-1}$. We conclude that each vector in the set

$$\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_j\} \quad (5)$$

is perpendicular to each of its predecessors. Thus the set (5) of vectors consists of j mutually perpendicular nonzero vectors in the j -dimensional subspace W_j , and so the set constitutes an orthogonal basis for W_j . It follows that, if we set $\mathbf{q}_i = (1/\|\mathbf{v}_i\|)\mathbf{v}_i$ for $i = 1, 2, \dots, j$, then $W_j = \text{sp}(\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_j)$. Taking $j = k$, we see that

$$\{\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_k\}$$

is an orthonormal basis for W . $\blacktriangle$

The proof of Theorem 6.4 was computational, providing us with a technique for constructing an orthonormal basis for a subspace W of $\mathbb{R}^n$. The technique is known as the *Gram-Schmidt process*, and we have boxed it for easy reference.

Gram-Schmidt Process

To find an orthonormal basis for a subspace W of $\mathbb{R}^n$:

1. Find a basis $\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_j\}$ for W .
2. Let $\mathbf{v}_1 = \mathbf{a}_1$. For $i = 2, \dots, j$, compute in succession the vector $\mathbf{v}_i$ given by subtracting from $\mathbf{a}_i$ its projection on the subspace generated by its predecessors.
3. The $\mathbf{v}_i$ so obtained form an orthogonal basis for W , and they may be normalized to yield an orthonormal basis.

When actually executing the Gram-Schmidt process, we project vectors on subspaces, as described in step 2 in the box. We know that it is best to work with an orthogonal or orthonormal basis for a subspace when projecting on it; and because the subspace $W_{j-1} = \text{sp}(\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_{j-1})$ is also the subspace generated by the orthogonal set $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{j-1}\}$, it is surely best to work with the latter basis for W_{j-1} when computing the desired projection of $\mathbf{a}_j$ on W_{j-1} . Step 2 in the box and Eq. (3) show that the specific formula for $\mathbf{v}_j$ is as follows:

General Gram-Schmidt Formula

$$\mathbf{v}_j = \mathbf{a}_j - \left(\frac{\mathbf{a}_j \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1} \mathbf{v}_1 + \frac{\mathbf{a}_j \cdot \mathbf{v}_2}{\mathbf{v}_2 \cdot \mathbf{v}_2} \mathbf{v}_2 + \cdots + \frac{\mathbf{a}_j \cdot \mathbf{v}_{j-1}}{\mathbf{v}_{j-1} \cdot \mathbf{v}_{j-1}} \mathbf{v}_{j-1} \right). \quad (6)$$

One may normalize the $\mathbf{v}_j$, forming the vector $\mathbf{q}_j = (1/\|\mathbf{v}_j\|)\mathbf{v}_j$, to obtain a vector of length 1 at each step of the construction. In that case, formula (6) can be replaced by the following simple form:

Normalized Gram-Schmidt Formula

$$\mathbf{v}_j = \mathbf{a}_j - ((\mathbf{a}_j \cdot \mathbf{q}_1)\mathbf{q}_1 + (\mathbf{a}_j \cdot \mathbf{q}_2)\mathbf{q}_2 + \cdots + (\mathbf{a}_j \cdot \mathbf{q}_{j-1})\mathbf{q}_{j-1}) \quad (7)$$

The arithmetic using formula (6) and that using formula (7) are similar, but formula (6) postpones the introduction of the radicals from normalizing until the entire orthogonal basis is obtained. We shall use formula (6) in our work. However, a computer will generate less error if it normalizes as it goes along. This is indicated in the next section.

EXAMPLE 4 Find an orthonormal basis for the subspace $W = \text{sp}([1, 0, 1], [1, 1, 1])$ of $\mathbb{R}^3$.

SOLUTION We use the Gram-Schmidt process with formula (6), finding first an orthogonal basis for W . We take $\mathbf{v}_1 = [1, 0, 1]$. From formula (6) with $\mathbf{v}_1 = [1, 0, 1]$ and $\mathbf{a}_2 = [1, 1, 1]$, we have

$$\mathbf{v}_2 = \mathbf{a}_2 - \frac{\mathbf{a}_2 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1} \mathbf{v}_1 = [1, 1, 1] - (2/2)[1, 0, 1] = [0, 1, 0].$$

An orthogonal basis for W is $\{[1, 0, 1], [0, 1, 0]\}$, and an orthonormal basis is $\{[1/\sqrt{2}, 0, 1/\sqrt{2}], [0, 1, 0]\}$. ■

Referring to the proof of Theorem 6.4, we see that the sets $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_j\}$ and $\{\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_j\}$ are both bases for the subspace $W_j = \text{sp}(\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_j)$. Consequently, the vector $\mathbf{a}_j$ can be expressed as a linear combination

$$\mathbf{a}_j = r_{1j}\mathbf{q}_1 + r_{2j}\mathbf{q}_2 + \cdots + r_{jj}\mathbf{q}_j. \quad (8)$$

HISTORICAL NOTE THE GRAM-SCHMIDT PROCESS is named for the Danish mathematician Jorgen P. Gram (1850–1916) and the German Erhard Schmidt (1876–1959). It was first published by Gram in 1883 in a paper entitled “Series Development Using the Method of Least Squares.” It was published again with a careful proof by Schmidt in 1907 in a work on integral equations. In fact, Schmidt even referred to Gram’s result. For Schmidt, as for Gram, the vectors were continuous functions defined on an interval $[a, b]$ with the inner product of two such functions ϕ, ψ being given as $\int_a^b \phi(x)\psi(x) dx$. Schmidt was more explicit than Gram, however, writing out the process in great detail and proving that the set of functions ϕ_i derived from his original set ϕ_i was in fact an orthonormal set.

Schmidt, who was at the University of Berlin from 1917 until his death, is best known for his definitive work on Hilbert spaces—spaces of square summable sequences of complex numbers. In fact, he applied the Gram-Schmidt process to sets of vectors in these spaces to help develop necessary and sufficient conditions for such sets to be linearly independent.

In particular, we see that $\mathbf{a}_1 = r_{11}\mathbf{q}_1$, $\mathbf{a}_2 = r_{12}\mathbf{q}_1 + r_{22}\mathbf{q}_2$, and so on. These equations arising from Eq. (8) for $j = 1, 2, \dots, k$ can be written in matrix form as

$$\begin{bmatrix} | & | & & | \\ \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_k \\ | & | & & | \end{bmatrix} = \begin{bmatrix} | & | & & | \\ \mathbf{q}_1 & \mathbf{q}_2 & \cdots & \mathbf{q}_k \\ | & | & & | \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & \cdots & r_{1k} \\ r_{22} & \cdots & & r_{2k} \\ \vdots & & \ddots & \vdots \\ 0 & & & r_{kk} \end{bmatrix},$$

A
 Q
 R

so $A = QR$ for the indicated matrices. Because each $\mathbf{a}_j$ is in W_j but not in W_{j-1} , we see that no r_{jj} is zero, so R is an invertible $k \times k$ matrix. This factorization $A = QR$ is important in numerical linear algebra; we state it as a corollary. We will find use for it in Sections 6.5 and 8.4.

COROLLARY 1 QR-Factorization

Let A be an $n \times k$ matrix with independent column vectors in $\mathbb{R}^n$. There exists an $n \times k$ matrix Q with orthonormal column vectors and an upper-triangular invertible $k \times k$ matrix R such that $A = QR$.

EXAMPLE 5 Let

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}.$$

Factor A in the form $A = QR$ described in Corollary 1 of Theorem 6.4, using the computations in Example 4.

SOLUTION From Example 4, we see that we can take

$$Q = \begin{bmatrix} 1/\sqrt{2} & 0 \\ 0 & 1 \\ 1/\sqrt{2} & 0 \end{bmatrix}$$

and solve $QR = A$ for the matrix R . That is, we solve the matrix equation

$$\begin{bmatrix} 1/\sqrt{2} & 0 \\ 0 & 1 \\ 1/\sqrt{2} & 0 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} \\ 0 & r_{22} \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$$

for the entries r_{11} , r_{12} , and r_{22} . This corresponds to two linear systems of three equations each, but by inspection we see that $r_{11} = \sqrt{2}$, $r_{12} = \sqrt{2}$, and $r_{22} = 1$. Thus,

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} & 0 \\ 0 & 1 \\ 1/\sqrt{2} & 0 \end{bmatrix} \begin{bmatrix} \sqrt{2} & \sqrt{2} \\ 0 & 1 \end{bmatrix} = QR.$$

We give another illustration of the Gram-Schmidt process, this time requiring two applications of formula (6).

EXAMPLE 6 Find an orthonormal basis for the subspace

$$W = \text{sp}([1, 2, 0, 2], [2, 1, 1, 1], [1, 0, 1, 1])$$

of $\mathbb{R}^4$.

SOLUTION First we find an orthogonal basis, using formula (6). We take $\mathbf{v}_1 = [1, 2, 0, 2]$ and compute $\mathbf{v}_2$ by subtracting from $\mathbf{a}_2 = [2, 1, 1, 1]$ its projection on $\mathbf{v}_1$:

$$\mathbf{v}_2 = \mathbf{a}_2 - \frac{\mathbf{a}_2 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1} \mathbf{v}_1 = [2, 1, 1, 1] - \frac{6}{9} [1, 2, 0, 2] = \left[\frac{4}{3}, -\frac{1}{3}, 1, -\frac{1}{3} \right].$$

To ease computations, we replace $\mathbf{v}_2$ by the parallel vector $3\mathbf{v}_2$, which serves just as well, obtaining $\mathbf{v}_2 = [4, -1, 3, -1]$. Finally, we subtract from $\mathbf{a}_3 = [1, 0, 1, 1]$ its projection on the subspace $\text{sp}(\mathbf{v}_1, \mathbf{v}_2)$, obtaining

$$\begin{aligned} \mathbf{v}_3 &= \mathbf{a}_3 - \frac{\mathbf{a}_3 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1} \mathbf{v}_1 - \frac{\mathbf{a}_3 \cdot \mathbf{v}_2}{\mathbf{v}_2 \cdot \mathbf{v}_2} \mathbf{v}_2 \\ &= [1, 0, 1, 1] - \frac{3}{9} [1, 2, 0, 2] - \frac{6}{27} [4, -1, 3, -1] \\ &= \left[-\frac{2}{9}, -\frac{4}{9}, \frac{3}{9}, \frac{5}{9} \right]. \end{aligned}$$

Replacing $\mathbf{v}_3$ by $9\mathbf{v}_3$, we see that

$$\{[1, 2, 0, 2], [4, -1, 3, -1], [-2, -4, 3, 5]\}$$

is an orthogonal basis for W . Normalizing each vector to length 1, we obtain

$$\left\{ \frac{1}{3} [1, 2, 0, 2], \frac{1}{3\sqrt{3}} [4, -1, 3, -1], \frac{1}{3\sqrt{6}} [-2, -4, 3, 5] \right\}$$

as an orthonormal basis for W . ■

As you can see, the arithmetic involved in the Gram-Schmidt process can be a bit tedious with pencil and paper, but it is very easy to implement the process on a computer.

We know that any independent set of vectors in $\mathbb{R}^n$ can be extended to a basis for $\mathbb{R}^n$. Using Theorem 6.4, we can prove a similar result for orthogonal sets.

COROLLARY 2 Expansion of an Orthogonal Set to an Orthogonal Basis

Every orthogonal set of vectors in a subspace W of $\mathbb{R}^n$ can be expanded if necessary to an orthogonal basis for W .

PROOF An orthogonal set $\{v_1, v_2, \dots, v_r\}$ of vectors in W is an independent set by Theorem 6.2, and can be expanded to a basis $\{v_1, \dots, v_r, a_1, \dots, a_s\}$ of W by Theorem 2.3. We apply the Gram-Schmidt process to this basis for W . Because the v_j are already mutually perpendicular, none of them will be changed by the Gram-Schmidt process, which thus yields an orthogonal basis containing the given vectors v_j for $j = 1, \dots, r$. $\blacktriangle$

EXAMPLE 7 Expand $\{[1, 1, 0], [1, -1, 1]\}$ to an orthogonal basis for $\mathbb{R}^3$, and then transform this to an orthonormal basis for $\mathbb{R}^3$.

SOLUTION First we expand the given set to a basis $\{a_1, a_2, a_3\}$ for $\mathbb{R}^3$. We take $a_1 = [1, 1, 0]$, $a_2 = [1, -1, 1]$, and $a_3 = [1, 0, 0]$, which we can see form a basis for $\mathbb{R}^3$. (See Theorem 2.3.)

Now we use the Gram-Schmidt process with formula (6). Because a_1 and a_2 are perpendicular, we let $v_1 = a_1 = [1, 1, 0]$ and $v_2 = a_2 = [1, -1, 1]$. From formula (6), we have

$$\begin{aligned} v_3 &= a_3 - \left(\frac{a_3 \cdot v_1}{v_1 \cdot v_1} v_1 + \frac{a_3 \cdot v_2}{v_2 \cdot v_2} v_2 \right) \\ &= [1, 0, 0] - \left(\frac{1}{2} [1, 1, 0] + \frac{1}{3} [1, -1, 1] \right) \\ &= [1, 0, 0] - \left[\frac{5}{6}, \frac{1}{6}, \frac{1}{3} \right] = \left[\frac{1}{6}, -\frac{1}{6}, -\frac{1}{3} \right]. \end{aligned}$$

Multiplying this vector by -6 , we replace v_3 by $[-1, 1, 2]$. Thus we have expanded the given set to an orthogonal basis

$$\{[1, 1, 0], [1, -1, 1], [-1, 1, 2]\}$$

of $\mathbb{R}^3$. Normalizing these vectors to unit length, we obtain

$$\left\{ \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0 \right\}, \left\{ \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right\}, \left\{ -\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}} \right\}$$

as an orthonormal basis. $\blacksquare$

The Gram-Schmidt Process in Inner-Product Spaces (Optional)

The results in this section easily extend to any inner-product space. We have the notions of an orthogonal set, an orthogonal basis, and an orthonormal basis, with essentially the same definitions given earlier. The Gram-Schmidt process is still valid. We conclude with an example.

EXAMPLE 8 Find an orthogonal basis for the subspace $\text{sp}(1, \sqrt{x}, x)$ of the vector space C_0 of continuous functions with domain $0 \leq x \leq 1$, where $\langle f, g \rangle = \int_0^1 f(x)g(x) dx$.

SOLUTION We let $v_1 = 1$ and compute

$$v_2 = \sqrt{x} - \frac{\langle \sqrt{x}, 1 \rangle}{\langle 1, 1 \rangle} 1 = \sqrt{x} - \frac{\int_0^1 \sqrt{x} dx}{\int_0^1 1 dx} = \sqrt{x} - \frac{2/3}{1} = \sqrt{x} - \frac{2}{3}.$$